*STPM Physics · Revision notes*

# PhysiQ

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# SEMESTER 3 NOTES

Oscillations, Waves & Modern Physics · Chapters 20–26

**7 chapters · 35 topics.** Definitions and laws, formula tables with symbols and units, tips to answer, and the Section A traps and tricks that cost marks in the multiple-choice paper.

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**Trap:** a wrong option that looks right — the slip examiners rely on.

**Trick:** a quick check or shortcut for the multiple-choice paper.

Section A · Semester 3

# SECTION A GUIDE

## BEFORE YOU START: THE 6-STEP SECTION A ROUTINE

- **Read the LAST line first** — know what is being asked (a value, a ratio, a direction, "which is NOT true") before reading the data.
- **Circle the trigger words**: NOT, only, always, never, increases, decreases, ratio, "per unit", peak / r.m.s., radius / diameter.
- **Convert units immediately**: $mm^2 \rightarrow \times 10^{-6}\ m^2$, $cm \rightarrow \times 10^{-2}\ m$, $\mu = 10^{-6}$, $n = 10^{-9}$, $p = 10^{-12}$, °C → K, minutes → seconds.
- **Estimate first**: do a rough order-of-magnitude calculation (powers of ten) — usually two options are wildly off and can be removed before you do any precise arithmetic.
- **Match the answer to the distractors**: examiners build wrong options from a forgotten square, a missing factor of 2, an inverted ratio, a wrong sign, or an un-converted unit. If your answer matches a "classic slip" value, re-check that step.
- **Time-box**: about 1½ minutes per question. Mark a hard one, move on, return later. Never leave a question blank — eliminate what you can and choose.

## UNIVERSAL TRICKS THAT WORK IN EVERY SEMESTER

**Trick: Ratio method** — for "if X is doubled/tripled, what happens to Y?", write $Y \propto$ (formula), then substitute the factors (e.g. $\frac{2}{3^2} = \frac{2}{9}$). Never calculate with real numbers when a ratio will do.

**Trick: Dimension / unit check** — if the unit of your final expression is not the unit asked for, it is wrong. Quick way to remove 1–2 options.

**Trick: Extreme-case check** — imagine $r \rightarrow \infty$, $t \rightarrow 0$ or $t \rightarrow \infty$. Does the option behave sensibly (e.g. a discharging capacitor must end at zero)?

**Trick: Direction/sign questions** — draw a 2-second sketch with axes. Do not trust your head for left-hand/right-hand rules or signs.

**Trick: "Which statement is correct" questions** — test each option separately and write ✓ / × next to it; one clearly false word (e.g. "always", "only") normally makes an option false.

**Trick: Graph questions** — decide first what the gradient and the area under the graph mean physically, then read the options.

## SEMESTER 3 — HOTSPOTS (OSCILLATIONS, WAVES, OPTICS, MODERN PHYSICS)

**Trap: SHM**: the period does not depend on amplitude; $a = -\omega^2 x$ (acceleration is maximum at the ends, speed is maximum at the centre); total energy $\propto A^2$.

**Trap: ω vs f again**: $\omega = 2\pi f = \frac{2\pi}{T}$. Read the equation $y = A\sin(\omega t)$ carefully for ω before finding f or T.

**Trap: Frequency stays the same when a wave changes medium**; v and λ change ($v = f\lambda$). In refraction, $\lambda \propto v$.

**Trap: Intensity $\propto A^2$**, and $I \propto \frac{1}{r^2}$ for a point source (so amplitude $\propto \frac{1}{r}$).

**Trap: Path difference vs phase difference**: $\varphi = \frac{2\pi}{\lambda} \times$ path difference. Constructive: path difference = nλ; destructive: (n + ½)λ — unless there is a phase change at reflection.

**Trap: Standing waves**: adjacent nodes are $\frac{\lambda}{2}$ apart. A pipe closed at one end supports ONLY odd harmonics (f, 3f, 5f…); an open pipe supports all harmonics.

**Trap: Decibels**: $\beta = 10 \log \frac{I}{I_0}$. Doubling the intensity adds about 3 dB; ×10 adds 10 dB. Intensity levels do NOT add arithmetically for two sources — add the intensities, then take the log.

**Trap: Doppler effect**: the observer moving TOWARDS the source or the source moving towards the observer raises f; get the signs of the velocities right with a sketch.

**Trap: Mirrors/lenses**: use ONE sign convention (real-is-positive). A virtual image has negative v; a negative magnification means inverted.

**Trap: Thin films**: check for a π phase change at each reflection (from a denser medium) BEFORE choosing $2nt = n\lambda$ or $(n + ½)\lambda$.

**Trap: Photoelectric effect**: intensity changes the NUMBER of photoelectrons, not their maximum kinetic energy; only frequency (above threshold) changes $E_k(\max)$.

**Trap: Radioactivity**: after n half-lives the remaining fraction is $(½)^n$ — not $1 - \frac{n}{2}$. Activity and the number of undecayed nuclei fall together.

**Trick: Two-slit**: fringe spacing $x = \frac{\lambda D}{a}$; doubling D or λ doubles x, doubling a halves x.

**Trick: Diffraction grating**: $d\sin\theta = n\lambda$. The maximum order is the largest integer $n \le \frac{d}{\lambda}$ (because $\sin\theta \le 1$).

**Trick: Photon energy**: $E = hf = \frac{hc}{\lambda} \approx \frac{1240}{\lambda(\text{nm})}$ eV. This single conversion removes lots of calculation.

**Trick: Bohr hydrogen**: $E_n = -\frac{13.6}{n^2}$ eV. Transitions to n = 1 give ultraviolet, to n = 2 visible, to n = 3 infrared.

**Trick: Mass defect**: 1 u ≈ 931.5 MeV. Binding energy per nucleon peaks near iron — fusion releases energy for light nuclei, fission for heavy ones.

**Trick: Malus's law**: $I = I_0 \cos^2\theta$; unpolarised light passing through a polariser is halved.

Chapter 20

# OSCILLATIONS

## 20.1 Characteristics of Simple Harmonic Motion

### DEFINITIONS & LAWS

- **Simple harmonic motion** is the motion of a particle whose acceleration is directly proportional to its displacement from a fixed (equilibrium) point and is always directed towards that point.
- **Amplitude** is the maximum displacement of the particle from the equilibrium position (m).
- **Equilibrium position** is the point at which no resultant force acts on the body performing SHM; displacement is measured from this point.
- **Period** is the time taken for one complete oscillation; **frequency** is the number of complete oscillations per unit time.
- **Angular frequency** is the rate of change of phase angle, $\omega = 2\pi f = \frac{2\pi}{T}$ (rad $s^{-1}$).
- **Phase difference** is the fraction of a cycle by which one oscillation is ahead of or behind another, expressed in radian or degree.
- In SHM the restoring force is directly proportional to the displacement and opposite in direction to it. Velocity leads displacement by $\frac{\pi}{2}$ rad; acceleration is in antiphase (π rad) with displacement.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Defining equation of SHM** | $a = -\omega^2 x$ | a = acceleration (m $s^{-2}$); ω = angular frequency (rad $s^{-1}$); x = displacement (m) |
| **Restoring force** | $F = -kx = -m\omega^2 x$ | k = force constant (N $m^{-1}$); m = mass (kg); x = displacement (m) |
| **Angular frequency** | $\omega = 2\pi f = \frac{2\pi}{T}$ | f = frequency (Hz); T = period (s) |
| **Maximum velocity** | $v_{max} = \omega A$ | At x = 0; A = amplitude (m) |
| **Maximum acceleration** | $a_{max} = \omega^2 A$ | At x = ±A; A = amplitude (m) |

- For a graph of a against x: a straight line through the origin with gradient $-\omega^2$.

### TIPS TO ANSWER

- Quote both conditions in a definition: acceleration ∝ displacement AND directed towards the equilibrium position (minus sign).
- Describe a graph of a against x as a straight line through the origin with negative gradient; the magnitude of the gradient is $\omega^2$.
- State the position of each extreme: at x = ±A, v = 0 and |a| and |F| are maximum; at x = 0, |v| is maximum and a = F = 0.
- For a phase relationship, give the angle with the direction: velocity leads displacement by $\frac{\pi}{2}$ rad.

### SECTION A TRAPS

**Trap:** Speed is maximum at the extreme positions. Wrong: the particle momentarily stops at the extremes; speed is maximum at the equilibrium position.

**Trap:** Reading the magnitude of the gradient of the a–x graph as ω or A. The gradient is $-\omega^2$, so $\omega = \sqrt{|\text{gradient}|}$.

**Trap:** Taking acceleration to be constant or the restoring force independent of displacement; both contradict $a = -\omega^2 x$.

### TRICKS

**Trick:** Velocity and acceleration are a quarter cycle apart in their extremes: when one is maximum, the other is zero.

**Trick:** Quick check on any SHM statement: if it says 'constant' or 'independent of displacement' for a or F, it is not SHM.

## 20.2 Kinematics of Simple Harmonic Motion

### DEFINITIONS & LAWS

- **Displacement** in SHM is a sinusoidal function of time; for a particle starting from the equilibrium position, $x = A \sin \omega t$.
- Velocity is the rate of change of displacement, $v = \frac{dx}{dt}$, and acceleration is the rate of change of velocity, $a = \frac{dv}{dt} = -\omega^2 x$.
- The speed at displacement x is given by $v = \pm\omega\sqrt{A^2 - x^2}$; the speed is zero at x = ±A and maximum at x = 0.
- The period of SHM is independent of the amplitude.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Displacement** | $x = A \sin \omega t$; $x = A \cos \omega t$ | A = amplitude (m); sine form starts at equilibrium; cosine form starts at the extreme position |
| **Velocity** | $v = \frac{dx}{dt} = \omega A \cos \omega t$ | v = velocity (m $s^{-1}$) |
| **Acceleration** | $a = -\omega^2 A \sin \omega t = -\omega^2 x$ | a = acceleration (m $s^{-2}$) |
| **Velocity at displacement x** | $v = \pm\omega\sqrt{A^2 - x^2}$ | x = displacement (m); A = amplitude (m) |
| **Maximum values and period** | $v_{max} = \omega A$; $a_{max} = \omega^2 A$; $T = \frac{2\pi}{\omega}$ | $v_{max}$ = maximum velocity (m $s^{-1}$); $a_{max}$ = maximum acceleration (m $s^{-2}$); T = period (s) |

- Calculator must be in radian mode when evaluating sin ωt and cos ωt.

## TIPS TO ANSWER

- Identify the starting condition first: x = 0 at t = 0 gives sine; x = A at t = 0 gives cosine.
- State formula, substitute with units, give the answer to 3 s.f. with unit (and direction when velocity is required).
- To find the time to reach a displacement, solve $\sin \omega t = \frac{x}{A}$ for ωt in radian, then $t = \frac{\omega t}{\omega}$.
- When the period is given, convert first: $\omega = \frac{2\pi}{T}$. When the amplitude is in cm, convert to m before using $v_{max} = \omega A$.

## SECTION A TRAPS

**Trap:** Using degree mode for ωt (ωt is in radian). The wrong mode gives a displacement and velocity far from the true values.

**Trap:** Using $v_{max} = \omega^2 A$ or $a_{max} = \omega A$. Check the units: ωA is in m $s^{-1}$, $\omega^2 A$ is in m $s^{-2}$.

**Trap:** Forgetting that $\omega = \frac{2\pi}{T}$, so quoting $v_{max} = \frac{A}{T}$ or using the frequency f in place of ω.

**Trap:** Using $v = \omega(A^2 - x^2)$ without the square root.

## TRICKS

**Trick:** Time to reach $\frac{A}{2}$ from equilibrium is $\frac{T}{12}$ (phase $\frac{\pi}{6}$); time to reach A is $\frac{T}{4}$.

**Trick:** Check v at x: $\frac{v}{v_{max}} = \sqrt{1 - (\frac{x}{A})^2}$. For x = 0.6A, v = 0.8$v_{max}$ (3-4-5 triangle).

# 20.3 Energy in Simple Harmonic Motion

## DEFINITIONS & LAWS

- **Total energy** of a particle in undamped SHM is constant and is equal to its maximum kinetic energy (at the equilibrium position) or its maximum potential energy (at the extreme positions).
- Kinetic energy is maximum at x = 0 and zero at x = ±A; potential energy is zero at x = 0 and maximum at x = ±A.
- Energy is continuously interchanged between kinetic and potential forms with the total remaining constant when there is no damping.
- The total energy is directly proportional to the square of the amplitude.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Total energy** | $E = \frac{1}{2}m\omega^2A^2 = \frac{1}{2}kA^2$ | m = mass (kg); ω = angular frequency (rad $s^{-1}$); A = amplitude (m); k = $m\omega^2$ (N $m^{-1}$) |
| **Kinetic energy** | $E_k = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(A^2 - x^2)$ | $E_k$ = kinetic energy (J); v = velocity (m $s^{-1}$); x = displacement (m) |
| **Potential energy** | $E_p = \frac{1}{2}m\omega^2x^2 = \frac{1}{2}kx^2$ | $E_p$ = potential energy (J); x = displacement (m) |
| **Conservation of energy** | $E_k + E_p = E$ | E is constant (J) |
| **Fractions of total energy** | $\frac{E_k}{E} = 1 - (\frac{x}{A})^2$; $\frac{E_p}{E} = (\frac{x}{A})^2$ | Ratios of kinetic and potential energy to the total energy; x = displacement (m); A = amplitude (m) |

## TIPS TO ANSWER

- Sketch of E against x: $E_p$ is a parabola with minimum at x = 0; $E_k$ is an inverted parabola; E is a horizontal line at $\frac{1}{2}kA^2$; the curves cross where $E_k = E_p$.
- Sketch of energy against time: $E_k$ and $E_p$ have frequency 2f (period $\frac{T}{2}$) and are in antiphase.
- State: $E = \frac{1}{2}kA^2$ → substitute with amplitude in m → answer in joule.
- For fractions, write $\frac{E_p}{E} = (\frac{x}{A})^2$ and substitute the ratio directly; no need for m or ω.

### SECTION A TRAPS

**Trap:** Taking energy to be proportional to A. The total energy is proportional to $A^2$: doubling the amplitude gives four times the energy.

**Trap:** Equal kinetic and potential energies at $x = \frac{A}{2}$. Wrong: $E_k = E_p$ when $x^2 = \frac{A^2}{2}$, i.e. $x = \frac{A}{\sqrt{2}} \approx 0.707A$.

**Trap:** Omitting the factor ½ in $E = \frac{1}{2}kA^2$, or using the displacement x instead of the amplitude A for the total energy.

**Trap:** Assuming $E_k = E_p$ at $x = \frac{A}{2}$ gives 50%: at $x = \frac{A}{2}$, $E_p$ = 25% and $E_k$ = 75% of the total.

### TRICKS

**Trick:** Use fractions: $x = \frac{A}{2}$ gives $E_p = \frac{1}{4}E$; $x = \frac{A}{4}$ gives $E_p = \frac{1}{16}E$; $x = \frac{A}{\sqrt{2}}$ gives $\frac{1}{2}E$.

**Trick:** Scaling: $E \propto A^2$ (same k), so tripling A gives 9 times the energy.

## 20.4 Systems in Simple Harmonic Motion

### DEFINITIONS & LAWS

- A **mass–spring system** performs SHM because the restoring force on the mass is directly proportional to its extension and is directed towards the equilibrium position (Hooke's law, $F = -kx$).
- A **simple pendulum** performs SHM only for small angular displacement ($\sin\theta \approx \theta$ in radian, up to about 10°); the restoring force along the arc is $mg\sin\theta \approx mg\theta$.
- The period of a simple pendulum is independent of the mass of the bob and of the amplitude (for small angles).
- The period of a mass–spring system is independent of the amplitude and, for a vertical spring, of g.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Mass–spring period** | $T = 2\pi\sqrt{\frac{m}{k}}$ | T = period (s); m = mass (kg); k = force constant (N $m^{-1}$) |

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Mass–spring angular frequency** | $\omega = \sqrt{\frac{k}{m}}$ | $\omega$ = angular frequency (rad $s^{-1}$); k = force constant (N $m^{-1}$); m = mass (kg) |
| **Simple pendulum period** | $T = 2\pi\sqrt{\frac{L}{g}}$ | L = length from the point of suspension to the centre of the bob (m); g = acceleration of free fall (m $s^{-2}$) |
| **Springs in parallel** | $k = k_1 + k_2$ | k = effective force constant (N $m^{-1}$); $k_1$, $k_2$ = force constants of the springs (N $m^{-1}$) |
| **Springs in series** | $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$ | k = effective force constant (N $m^{-1}$); $k_1$, $k_2$ = force constants of the springs (N $m^{-1}$) |
| **Proportionalities for period** | $T \propto \sqrt{m}$; $T \propto \sqrt{L}$; $T \propto \frac{1}{\sqrt{g}}$; $T \propto \frac{1}{\sqrt{k}}$ | T = period (s); m = mass (kg); L = length (m); g = acceleration of free fall (m $s^{-2}$); k = force constant (N $m^{-1}$) |

- g = 9.81 m $s^{-2}$ on Earth.

## TIPS TO ANSWER

- State the condition for SHM: small angle (pendulum), or Hooke's law obeyed (spring).
- Use the ratio method for changes: $\frac{T_2}{T_1} = \sqrt{\frac{m_2}{m_1}}$ or $\sqrt{\frac{L_2}{L_1}}$ or $\sqrt{\frac{g_1}{g_2}}$; show the ratio before the final answer.
- Derive the period by showing $a = -\frac{k}{m}x$, then compare with $a = -\omega^2 x$ to give $\omega^2 = \frac{k}{m}$.
- Give the period in seconds to 3 s.f.; if frequency is asked, take $f = \frac{1}{T}$.

## SECTION A TRAPS

**Trap:** Writing $T = 2\pi\sqrt{\frac{g}{L}}$ or $T = 2\pi\sqrt{\frac{k}{m}}$: inverted ratios. Check the unit: $\sqrt{\frac{m}{k}}$ and $\sqrt{\frac{L}{g}}$ both give seconds.

**Trap:** Believing a heavier bob lengthens the pendulum period. The pendulum period is independent of the mass.

**Trap:** Using the mass-spring result with the wrong sense: quadrupling the mass doubles the period ($T \propto \sqrt{m}$), it does not quadruple it.

**Trap:** On the Moon (smaller g) the pendulum period is longer, not shorter, since $T \propto \frac{1}{\sqrt{g}}$.

### TRICKS

**Trick:** Quadruple the length or mass: period doubles. A period ratio of 2 means a length ratio of 4.

**Trick:** Springs in parallel make the system stiffer, so the period decreases; in series the period increases.

## 20.5 Damped Oscillation

### DEFINITIONS & LAWS

- **Damped oscillation** is an oscillation in which the amplitude decreases with time because energy is dissipated (as heat) by resistive forces such as air resistance or friction.
- **Light (under) damping**: the system oscillates with amplitude decreasing exponentially with time; the period is slightly longer than that of the free oscillation (almost unchanged when the damping is light).
- **Critical damping**: the system returns to the equilibrium position in the shortest possible time without oscillating.
- **Heavy (over) damping**: the system returns to the equilibrium position slowly without oscillating.
- Explain damping: the resistive force does negative work on the oscillator, so the total mechanical energy decreases and the amplitude decreases.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Amplitude decay** | $A = A_0e^{-bt}$ | $A_0$ = initial amplitude (m); b = damping constant ($s^{-1}$); t = time (s) |
| **Energy decay** | $E \propto A^2$; $E = E_0e^{-2bt}$ | E = energy (J); $E_0$ = initial energy (J); b = damping constant ($s^{-1}$); t = time (s) |
| **Exponential decay of amplitude** | $\frac{A_2}{A_1} = \frac{A_1}{A_0}$ | For equal time intervals the amplitude falls by the same factor (exponential decay); $A_0$, $A_1$, $A_2$ = amplitudes at successive equal intervals (m) |
| **Half-life of amplitude** | $t½ = \frac{\ln 2}{b}$ | t½ = half-life of amplitude (s); b = damping constant ($s^{-1}$) |

### TIPS TO ANSWER

- For a displacement–time graph of light damping, sketch oscillations whose peaks lie on an exponentially decaying envelope; label both axes.

- For critical and heavy damping, sketch a curve that falls to zero without crossing the time axis; critical damping reaches zero faster than heavy damping.
- Always state the energy change: mechanical energy is converted to thermal energy by the resistive force.
- Give a use of each type: critical damping in a car suspension, a galvanometer and door closers.

### SECTION A TRAPS

**Trap:** Stating that the total energy stays constant while only the amplitude changes. Energy decreases as $A^2$ decreases.

**Trap:** Assuming the amplitude falls by a constant amount each cycle. It falls by a constant fraction (exponential).

**Trap:** Saying that heavy damping returns the system to equilibrium fastest. Critical damping is the fastest without oscillation.

**Trap:** Saying that damping greatly increases the frequency; for light damping the period is almost unchanged.

### TRICKS

**Trick:** Equal ratios: 0.12 → 0.075 gives a ratio of 0.625, so the next equal interval gives 0.075 × 0.625.

**Trick:** Halving every $t_1$ seconds: after n intervals the amplitude is $\frac{A_0}{2^n}$; the energy is $\frac{{A_0}^2}{4^n}$.

## 20.6 Forced Oscillation and Resonance

### DEFINITIONS & LAWS

- **Free oscillation** is the oscillation of a system without any external force acting on it after the initial disturbance; no energy is lost, so the amplitude is constant and the system oscillates at its natural frequency.
- **Forced oscillation** is an oscillation of a system driven by an external periodic force; it oscillates at the frequency of the driving force.
- **Resonance** occurs when the driving frequency is equal to the natural frequency of the system, giving maximum amplitude of oscillation (maximum energy transfer to the system).
- Effect of damping: increasing damping reduces the amplitude at resonance and makes the resonance curve broader (less sharp); the peak shifts slightly to lower frequency.
- At resonance the driving force leads the displacement by $\frac{\pi}{2}$ rad.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Natural frequency, mass–spring** | $f_0 = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$ | $f_0$ = natural frequency (Hz); k = force constant (N $m^{-1}$); m = mass (kg) |
| **Natural frequency, simple pendulum** | $f_0 = \frac{1}{2\pi}\sqrt{\frac{g}{L}}$ | $f_0$ = natural frequency (Hz); g = acceleration of free fall (m $s^{-2}$); L = length (m) |
| **Resonance condition** | $f_{driving} = f_0$ | $f_{driving}$ = frequency of the driving force (Hz); $f_0$ = natural frequency (Hz) |

- Graph: amplitude against driving frequency, with a peak at $f_0$.

## TIPS TO ANSWER

- For the resonance graph, label axes (amplitude against driving frequency), draw a peak at $f_0$ and show a second curve for heavier damping which is lower and broader.
- Explain resonance in two points: driving frequency equals natural frequency; energy transferred to the system at the maximum rate, so the amplitude is maximum.
- Give examples with a consequence: collapse of a bridge or a building in an earthquake; useful: tuning of a radio circuit, microwave oven.
- Calculation: find $f_0 = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$ and state that resonance occurs at this driving frequency.

## SECTION A TRAPS

**Trap:** Leaving out the 2π, i.e. quoting $\sqrt{\frac{k}{m}} = 22$ as a frequency in Hz. This is ω in rad $s^{-1}$; $f = \frac{\omega}{2\pi}$.

**Trap:** Believing that more damping gives a higher and sharper resonance peak. More damping gives a lower, broader peak.

**Trap:** Saying that resonance occurs at the largest driving force or the largest driving frequency; it depends on the match with $f_0$.

**Trap:** Confusing resonance with beats or interference.

## TRICKS

**Trick:** If options include $\sqrt{\frac{k}{m}}$ and $\sqrt{\frac{k}{m}}/2\pi$, the frequency in Hz is the one divided by 2π; the other is ω.

> **Trick:** Limiting case: no damping gives an infinite peak; heavy damping gives no distinct peak.

Chapter 21

# WAVE MOTION

## 21.1 Progressive Waves

### DEFINITIONS & LAWS

- A **progressive (travelling) wave** is a wave whose profile moves along the medium; it transfers energy from one point to another through a medium (or vacuum) by oscillations of the particles (or fields), without transferring matter.
- **Wavelength** is the distance between two successive points that are in phase on the wave (the distance travelled by the wave in one period).
- A **wavefront** is the locus of points which vibrate in phase.
- **Amplitude** is the maximum displacement of a particle from its equilibrium position; **period** is the time for one complete oscillation of a particle.
- **Wave speed** is the distance travelled by the wavefront per unit time.
- **Phase difference** between two points on a progressive wave depends on their separation along the direction of travel.
- In a transverse wave the oscillations are perpendicular to the direction of propagation; in a longitudinal wave they are parallel to it.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Wave speed** | $v = f\lambda$ | $v$ = wave speed (m $s^{-1}$); $f$ = frequency (Hz); $\lambda$ = wavelength (m) |
| **Wave travelling in +x direction** | $y = A\sin(\omega t - kx)$ | $y$ = displacement (m); $A$ = amplitude (m); $k = \frac{2\pi}{\lambda}$ = wave number (rad $m^{-1}$); $\omega = 2\pi f$ (rad $s^{-1}$) |
| **Wave speed in terms of ω and k** | $v = \frac{\omega}{k} = \frac{\lambda}{T}; f = \frac{\omega}{2\pi}$ | $v$ = wave speed (m $s^{-1}$); $\omega$ = angular frequency (rad $s^{-1}$); $k$ = wave number (rad $m^{-1}$); $T$ = period (s); $f$ = frequency (Hz) |
| **Phase difference** | $\varphi = \frac{2\pi}{\lambda}\Delta x; \varphi = \frac{2\pi\Delta t}{T}$ | $\varphi$ = phase difference (rad); $\Delta x$ = path separation (m); $\Delta t$ = time difference (s); $T$ = period (s) |
| **Wave travelling in −x direction** | $y = A\sin(\omega t + kx)$ | Travels in the −x direction; symbols as above |

### TIPS TO ANSWER

- Read ω and k from the equation by comparison with y = A sin(ωt − kx); then v = $\frac{\omega}{k}$ and f = $\frac{\omega}{2\pi}$.
- Phase difference: write φ = $\frac{2\pi\Delta x}{\lambda}$, substitute, and give the answer in radian (and degree if asked) with 3 s.f.
- For graphs: y against x gives λ and A; y against t gives T and A. Label axes with units.
- Say that energy is transferred in the direction of wave propagation but the particles only oscillate about their equilibrium positions.

### SECTION A TRAPS

**Trap:** Using v = ωk instead of $\frac{\omega}{k}$. Check units: $\frac{\omega}{k}$ gives m $s^{-1}$.

**Trap:** Taking the number multiplying t as f. In y = A sin(50t − 4.0x), 50 is ω (rad $s^{-1}$), so f = $\frac{50}{2\pi}$ ≈ 8.0 Hz.

**Trap:** Phase difference without 2π: φ = $\frac{\Delta x}{\lambda}$ is the fraction of a cycle, not the phase angle in radian.

**Trap:** Reading y = A sin[2π($\frac{t}{T}$ − $\frac{x}{\lambda}$)] with the denominators swapped; $\frac{t}{T}$ gives period and $\frac{x}{\lambda}$ gives wavelength.

### TRICKS

**Trick:** Two points separated by one wavelength are in phase (2π); separated by $\frac{\lambda}{2}$ they are in antiphase (π); by $\frac{\lambda}{4}$ the phase difference is $\frac{\pi}{2}$.

**Trick:** In the form sin[2π($\frac{t}{T}$ − $\frac{x}{\lambda}$)], read T and λ directly and take v = $\frac{\lambda}{T}$.

## 21.2 Wave Intensity

### DEFINITIONS & LAWS

- **Intensity** of a wave is the rate of energy transferred per unit area perpendicular to the direction of propagation (power per unit area), unit W $m^{-2}$.
- Intensity is directly proportional to the square of the amplitude (for the same frequency and medium).
- For a point source radiating uniformly in all directions with no absorption, intensity is inversely proportional to the square of the distance from the source (inverse square law).

- Explain the inverse square law: the same power spreads over the surface of a sphere of area $4\pi r^2$, which increases as $r^2$.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Intensity** | $I = \frac{P}{A}$ | I = intensity (W $m^{-2}$); P = power (W); A = area normal to the wave ($m^2$) |
| **Intensity of a point source** | $I = \frac{P}{4\pi r^2}$ | Point source; r = distance from the source (m); P = power (W) |
| **Intensity and amplitude** | $I \propto A^2$; $\frac{I_2}{I_1} = (\frac{A_2}{A_1})^2$ | A = amplitude; I = intensity (W $m^{-2}$) |
| **Inverse square law** | $I_1 r_1^2 = I_2 r_2^2$; $I_2 = I_1(\frac{r_1}{r_2})^2$ | $I_1$, $I_2$ = intensities at distances $r_1$, $r_2$ from the source (W $m^{-2}$, m) |
| **Amplitude of a spherical wave** | $A \propto \frac{1}{r}$ | Spherical wave; A = amplitude; r = distance from the source (m) |

## TIPS TO ANSWER

- Choose between $I \propto \frac{1}{r^2}$ (distance change) and $I \propto A^2$ (amplitude change), and write the relation before substituting.
- Show the ratio: $I_2 = I_1(\frac{r_1}{r_2})^2$ with values and unit, then the answer to 3 s.f. in W $m^{-2}$.
- State the assumptions: point source, spreads uniformly, no absorption by the medium.
- For power calculations, use area $4\pi r^2$ and give the answer with unit W $m^{-2}$.

## SECTION A TRAPS

**Trap:** Using $I \propto \frac{1}{r}$ instead of $\frac{1}{r^2}$. Doubling the distance gives one quarter, not one half.

**Trap:** Inverting the ratio: moving farther must decrease the intensity, so the factor $(\frac{r_1}{r_2})^2$ is less than 1.

**Trap:** Tripling the amplitude triples the intensity. Wrong: $I \propto A^2$, so the intensity becomes 9 times.

**Trap:** Using the diameter as r in $I = \frac{P}{4\pi r^2}$.

## TRICKS

**Trick:** Distance × 3 → intensity ÷ 9; amplitude × 3 → intensity × 9.

**Trick:** Intensity down by a factor of 4 means distance up by a factor of 2: $r_2 = r_1\sqrt{\frac{I_1}{I_2}}$.

# 21.3 Principle of Superposition

## DEFINITIONS & LAWS

- **Principle of superposition**: when two or more waves meet at a point, the resultant displacement is the vector sum of the displacements of the individual waves at that point.
- **Coherent sources** are sources that emit waves of the same frequency with a constant phase difference.
- **Constructive interference** occurs when the waves meet in phase: the path difference is nλ (n = 0, 1, 2, ...), giving the maximum resultant amplitude.
- **Destructive interference** occurs when the waves meet in antiphase: the path difference is (n + ½)λ, giving the minimum resultant amplitude.
- A stable, observable interference pattern requires coherent waves of similar amplitude and the same frequency.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Constructive interference** | $\Delta x = n\lambda$ | $\Delta x$ = path difference (m); $\lambda$ = wavelength (m); n = 0, 1, 2, ...; coherent in-phase sources |
| **Destructive interference** | $\Delta x = (n + ½)\lambda$ | $\Delta x$ = path difference (m); $\lambda$ = wavelength (m); n = 0, 1, 2, ...; coherent in-phase sources |
| **Phase difference** | $\varphi = \frac{2\pi}{\lambda}\Delta x$ | $\varphi$ = phase difference (rad); $\Delta x$ = path difference (m); $\lambda$ = wavelength (m) |
| **Resultant amplitude** | $A^2 = A_1^2 + A_2^2 + 2A_1A_2\cos\varphi$ | Two waves of amplitudes $A_1$ and $A_2$ (m); $\varphi$ = phase difference (rad) |
| **Intensity of equal waves** | $I = 4I_0\cos^2\frac{\varphi}{2}$ | Equal waves; $I_0$ = intensity of each wave (W $m^{-2}$); $\varphi$ = phase difference (rad) |
| **Special cases of phase difference** | $\varphi = \frac{\pi}{2}$: $A = \sqrt{A_1^2 + A_2^2}$; $\varphi = 0$: $A = A_1 + A_2$; $\varphi = \pi$: $A = \lvert A_1 - A_2\rvert$ | A = resultant amplitude (m); $A_1$, $A_2$ = amplitudes of the two waves (m); $\varphi$ in rad |

### TIPS TO ANSWER

- Compute $\frac{\Delta x}{\lambda}$ and state the result: a whole number gives constructive; an odd number of half-wavelengths gives destructive.
- Define coherence completely: same frequency and constant phase difference (not just 'same amplitude').
- For the amplitude of the resultant wave, state φ, then substitute into the amplitude relation with unit.
- Compare the intensity with $I_0$ using $I \propto A^2$, and state the answer in terms of $I_0$.

### SECTION A TRAPS

**Trap:** Concluding destructive interference just because the path difference is 'not an integer'. Check whether $\frac{\Delta x}{\lambda}$ is a half-integer, e.g. 1.5.

**Trap:** Simply adding amplitudes (3 + 4 = 7) for a phase difference of $\frac{\pi}{2}$; the correct result is $\sqrt{3^2 + 4^2}$ = 5.

**Trap:** Converting the path difference to phase without 2π: $\varphi = \frac{2\pi\Delta x}{\lambda}$, so $\Delta x = \frac{\lambda}{3}$ gives $\frac{2\pi}{3}$ rad.

**Trap:** Saying that waves of different frequencies give a stable pattern; a constant phase difference requires the same frequency.

### TRICKS

**Trick:** $\frac{\Delta x}{\lambda}$ = 3.0, 1.5, 0.75: integer → bright (constructive); half-integer → dark (destructive); anything else is in between.

**Trick:** $I = 4I_0\cos^2\frac{\varphi}{2}$: $\varphi = \frac{2\pi}{3}$ gives $I_0$; φ = π gives 0; φ = 0 gives $4I_0$.

## 21.4 Standing Waves

### DEFINITIONS & LAWS

- A **standing (stationary) wave** is a wave whose profile does not move; it is formed by the superposition of two progressive waves of the same frequency and amplitude travelling in opposite directions.
- A **node** is a point of zero amplitude (permanent destructive interference); an **antinode** is a point of maximum amplitude.

- Adjacent nodes (or adjacent antinodes) are separated by $\frac{\lambda}{2}$; a node and the adjacent antinode are separated by $\frac{\lambda}{4}$.
- Differences from a progressive wave: a standing wave does not transfer energy along the medium; its amplitude varies with position (zero at nodes); particles between two adjacent nodes oscillate in phase, and particles in adjacent segments oscillate in antiphase.
- For a string fixed at both ends there is a node at each end; only wavelengths with an integral number of half-wavelengths fit on the string.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Standing wave on string fixed at both ends** | $L = \frac{n\lambda}{2}$; $f_n = \frac{nv}{2L}$ | n = 1, 2, 3, ...; v = wave speed on the string (m $s^{-1}$); L = length (m); λ = wavelength (m); $f_n$ = frequency of the nth mode (Hz) |
| **Fundamental and harmonics** | $f_1 = \frac{v}{2L}$; $f_n = nf_1$ | $f_1$ = fundamental frequency (Hz); $f_n$ = frequency of the nth harmonic (Hz); v = wave speed (m $s^{-1}$); L = length (m) |
| **Standing wave equation** | $y = 2A \sin(kx) \cos(\omega t)$ | $k = \frac{2\pi}{\lambda}$ (rad $m^{-1}$); A = amplitude of each travelling wave (m); ω = angular frequency (rad $s^{-1}$); nodes at $kx = n\pi$ |
| **Node spacing** | Distance between adjacent nodes = $\frac{\lambda}{2} = \frac{\pi}{k}$ | λ = wavelength (m); $k = \frac{2\pi}{\lambda}$ = wave number (rad $m^{-1}$) |
| **Wave speed on a string** | $v = \sqrt{\frac{T}{\mu}}$ | Stretched string; T = tension (N); μ = mass per unit length (kg $m^{-1}$) |

## TIPS TO ANSWER

- Sketch the pattern with N at both ends for a string and label the nodes, antinodes and the wavelength on the diagram.
- State the condition $L = \frac{n\lambda}{2}$, then $f = \frac{v}{\lambda} = \frac{nv}{2L}$; give the answer to 3 s.f. in Hz.
- In an 'explain' question state: formed by superposition of the incident and reflected waves of the same frequency and amplitude travelling in opposite directions.
- From $y = 2A \sin(kx) \cos(\omega t)$ identify k, then $\lambda = \frac{2\pi}{k}$ and the node separation $\frac{\lambda}{2}$.

## SECTION A TRAPS

**Trap:** Taking the node separation to be λ rather than $\frac{\lambda}{2}$.

**Trap:** Using $f_1 = \frac{v}{L}$ for a string fixed at both ends; the fundamental has $L = \frac{\lambda}{2}$, so $f_1 = \frac{v}{2L}$.

**Trap:** Counting the gaps between nodes wrongly: from the 2nd node to the 5th node there are three gaps, i.e. $3 \times \frac{\lambda}{2}$.

**Trap:** Saying that a standing wave transports energy along the string, or that all points have the same amplitude.

### TRICKS

**Trick:** The nth harmonic is n times the fundamental; the third harmonic of a 200 Hz fundamental is 600 Hz.

**Trick:** From the number of gaps g between two nodes, distance = $g\frac{\lambda}{2}$; from a node at 0 and k, the spacing is $\frac{\pi}{k}$.

## 21.5 Electromagnetic Waves

### DEFINITIONS & LAWS

- **Electromagnetic waves** are transverse waves consisting of oscillating electric and magnetic fields that are perpendicular to each other and to the direction of propagation; they do not need a material medium.
- All electromagnetic waves travel at the same speed, $c = 3.00\times10^8$ m $s^{-1}$, in vacuum; they differ in frequency and wavelength.
- In an electromagnetic wave, the electric and magnetic fields oscillate in phase.
- Order of the spectrum, from longest wavelength: radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, gamma rays.
- Electromagnetic waves can be polarised, reflected, refracted, diffracted and show interference (properties of transverse waves).

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Wave equation for EM waves** | $c = f\lambda$ | $c$ = speed of light in vacuum ($3.00\times10^{8}$ m s$^{-1}$); $f$ = frequency (Hz); $\lambda$ = wavelength (m) |
| **Field amplitudes** | $E_0 = cB_0$ | $E_0$ = electric field amplitude (V m$^{-1}$); $B_0$ = magnetic field amplitude (T); $c$ = speed of light (m s$^{-1}$) |
| **Speed of light** | $c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$ | $\mu_0 = 4\pi\times10^{-7}$ H m$^{-1}$; $\varepsilon_0 = 8.85\times10^{-12}$ F m$^{-1}$ |
| **Photon energy** | $E = hf = \frac{hc}{\lambda}$ | $E$ = photon energy (J); $h = 6.63\times10^{-34}$ J s; $f$ = frequency (Hz); $\lambda$ = wavelength (m) |
| **Time of travel** | $t = \frac{d}{c}$ | $t$ = time (s); $d$ = distance (m); $c$ = speed of light (m s$^{-1}$) |

## TIPS TO ANSWER

- State the nature: transverse; E and B perpendicular to each other and to the direction of propagation; in phase.
- For calculations: formula → substitution with $c = 3.00\times10^{8}$ m s$^{-1}$ → answer to 3 s.f. with unit and in standard form.
- For the spectrum, order by wavelength or frequency and state the direction, e.g. 'increasing frequency'.
- State that no medium is required, so EM waves travel through vacuum.

## SECTION A TRAPS

**Trap:** Reversing the order of the spectrum: radio waves have the longest wavelength and lowest frequency; X-rays are shorter than ultraviolet.

**Trap:** Using $B = cE$ instead of $B = \frac{E}{c}$, which gives an absurd $10^{11}$ T; B is of the order of $10^{-6}$ T for E of a few hundred V m$^{-1}$.

**Trap:** Saying that E and B are 90° out of phase. They are in phase (both reach maxima together).

**Trap:** Saying that EM waves are longitudinal or that they need a medium.

## TRICKS

**Trick:** Quick check: $\lambda = \frac{c}{f} = 3\times\frac{10^{8}}{1.5}\times10^{14} = 2\times10^{-6}$ m; always examine the power of ten against the spectrum.

> **Trick:** $\frac{E}{B}$ must equal $3\times10^{8}$ m $s^{-1}$ in vacuum; use it to check magnitudes in options.

Chapter 22

# SOUND WAVES

## 22.1 Propagation of Sound Waves

### DEFINITIONS & LAWS

- **Sound waves** are longitudinal mechanical waves in which the particles of the medium oscillate parallel to the direction of propagation, forming alternate regions of compression and rarefaction.
- A sound wave needs a material medium; it cannot travel through a vacuum, and there is no net flow of the medium from source to listener.
- The **speed of sound** depends on the elasticity and density of the medium: $v = \sqrt{\frac{B}{\rho}}$ in a fluid and $v = \sqrt{\frac{E}{\rho}}$ in a solid, so v is greater in solids than in liquids and gases.
- In an ideal gas $v = \sqrt{\frac{\gamma RT}{M}} = \sqrt{\frac{\gamma p}{\rho}}$, so v is proportional to $\sqrt{T}$ (T in kelvin) and independent of pressure at constant temperature.
- **Intensity** is the rate of energy flow per unit area perpendicular to the direction of propagation; $I = \frac{P}{A}$ and $I \propto (\text{amplitude})^2$.
- Displacement nodes are pressure antinodes and displacement antinodes are pressure nodes; the pressure variation is $\frac{\pi}{2}$ out of phase with the displacement.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Wave speed** | $v = f\lambda$ | v = wave speed (m $s^{-1}$); f = frequency (Hz); λ = wavelength (m) |
| **Sound wave equation** | $y = A \sin(\omega t - kx)$ | $\omega = 2\pi f$ (rad $s^{-1}$); $k = \frac{2\pi}{\lambda}$ (rad $m^{-1}$); wave speed $v = \frac{\omega}{k}$ (m $s^{-1}$); A = amplitude |
| **Speed of sound and temperature** | $\frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}}$ | T = absolute temperature (K); v = speed of sound (m $s^{-1}$); $v \approx 331$ m $s^{-1}$ at 0 °C |
| **Intensity of a point source** | $I = \frac{P}{A} = \frac{P}{4\pi r^2}$; $\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2}$ | Point source radiating uniformly; I = intensity (W $m^{-2}$); P = power (W); A = area ($m^2$); r = distance from the source (m) |

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Echo / sonar distance** | $d = \frac{vt}{2}$ | d = distance to the reflector (m); v = speed of sound (m $s^{-1}$); t = total time for the pulse to travel to the reflector and back (s) |

## TIPS TO ANSWER

- State that sound is a longitudinal wave and name compressions and rarefactions; mention the direction of particle vibration relative to the direction of travel.
- For $y = A\sin(\omega t - kx)$, read off ω and k by comparison, then calculate $v = \frac{\omega}{k}$ and $f = \frac{\omega}{2\pi}$; include units.
- To explain why v differs between media, state that v depends on (elastic modulus)/(density), not on density alone.
- For temperature problems, convert °C to kelvin before taking the ratio of square roots; state the answer to 3 s.f.

## SECTION A TRAPS

**Trap:** Using Celsius temperatures in $v \propto \sqrt{T}$. The ratio must be of absolute temperatures, e.g. $\sqrt{\frac{303}{273}}$, not $\sqrt{\frac{30}{0}}$.

**Trap:** Using $v = \frac{\omega}{k}$ with k replaced by λ, or $f = \omega$ instead of $f = \frac{\omega}{2\pi}$; this gives 300 Hz instead of 47.7 Hz.

**Trap:** Saying a denser medium always carries sound faster. Steel is denser than air but has a far larger elastic modulus, so v is greater.

**Trap:** Forgetting the factor ½ in echo and sonar problems: the pulse travels to the reflector and back.

## TRICKS

**Trick:** Speed of sound ratio: a 30 K rise near room temperature changes v by only about 5%, so any option far from 331 × 1.05 can be rejected at once.

**Trick:** In $y = A\sin(\omega t - kx)$, $v = \frac{\text{coefficient of } t}{\text{coefficient of } x}$; no need to find f first.

# 22.2 Sources of Sound

## DEFINITIONS & LAWS

- **Standing (stationary) waves** in air columns and strings are formed by the superposition of a progressive wave and its reflection, giving fixed nodes and antinodes.
- **Resonance** occurs when the driving frequency equals a natural frequency of the system, producing a large-amplitude oscillation.
- In an **open pipe** there is a displacement antinode at each end, so all harmonics are present: $f_n = \frac{nv}{2L}$.
- In a **closed pipe** there is a displacement node at the closed end and an antinode at the open end, so only odd harmonics are present: $f_n = \frac{(2n - 1)v}{4L}$.
- In a stretched string fixed at both ends there is a node at each end; the wave speed on the string is $v = \sqrt{\frac{T}{\mu}}$ and $f_n = \frac{nv}{2L}$.
- The antinode at an open end lies slightly beyond the pipe; this **end correction** makes the effective length of the air column larger than the physical length.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Open pipe harmonics** | $f_n = \frac{nv}{2L}$; $\lambda_n = \frac{2L}{n}$ | $f_n$ = frequency of nth harmonic (Hz); n = 1, 2, 3, …; v = speed of sound (m $s^{-1}$); L = length of pipe (m); $\lambda_n$ = wavelength (m); pipe open at both ends |
| **Closed pipe harmonics** | $f = \frac{v}{4L}, \frac{3v}{4L}, \frac{5v}{4L}, \ldots$; $\lambda_1 = 4L$ | f = frequency (Hz); v = speed of sound (m $s^{-1}$); L = length of pipe (m); $\lambda_1$ = fundamental wavelength (m); pipe closed at one end |
| **Stretched string harmonics** | $f_n = \frac{n}{2L}\sqrt{\frac{T}{\mu}}$ | $f_n$ = frequency (Hz); n = 1, 2, 3, …; L = length of string (m); T = tension (N); μ = mass per unit length (kg $m^{-1}$) |
| **Closed pipe fundamental** | $v = 4fL$; $L + e = \frac{\lambda}{4}$ | v = speed of sound (m $s^{-1}$); f = frequency (Hz); L = length of air column at first resonance (m); e = end correction (m); λ = wavelength (m) |

- Speed of sound in air ≈ 340 m $s^{-1}$ unless otherwise stated.

## TIPS TO ANSWER

- Draw the displacement pattern (node/antinode) for the pipe first, then write L in terms of λ: open $L = \frac{n\lambda}{2}$, closed $L = \frac{(2n - 1)\lambda}{4}$.
- Identify the harmonic number by dividing the given frequency by the fundamental; for a closed pipe it must be an odd integer.

- State formula, substitute with L in metres, and give the frequency in Hz to 3 s.f.
- For 'second lowest resonance' of a closed pipe, the answer is $3f_1$, not $2f_1$.

### SECTION A TRAPS

**Trap:** Applying $f = \frac{nv}{2L}$ to a closed pipe, or $f = \frac{v}{4L}$ to an open pipe. The fundamental of a closed pipe is half that of an open pipe of the same length.

**Trap:** Counting the 'second harmonic' of a closed pipe as $2f_1$. Even multiples are absent; the next resonance is $3f_1$.

**Trap:** Using the diameter or the pipe length in cm without conversion: L must be in metres.

**Trap:** Treating the first overtone and the second harmonic as different. For an open pipe the first overtone is the second harmonic ($2f_1$).

### TRICKS

**Trick:** Ratio method: closed fundamental = ½ × open fundamental for equal L and v.

**Trick:** Frequency ratio test: if $\frac{f}{f_1}$ is an even integer or non-integer, the closed-pipe harmonic is impossible.

## 22.3 Intensity Level of Sound

### DEFINITIONS & LAWS

- **Intensity level** (β) is a logarithmic measure of the intensity of a sound relative to the threshold of hearing: $\beta = 10 \log_{10} \frac{I}{I_0}$, measured in decibels (dB).
- The **threshold of hearing** $I_0 = 1.0 \times 10^{-12}$ W m$^{-2}$ is the minimum intensity audible at about 1 kHz.
- A difference of 10 dB corresponds to a ratio of intensities of 10; a difference of about 3 dB corresponds to a ratio of 2.
- Intensity levels in dB cannot be added directly; the intensities of incoherent sources add, and the total is then converted to dB.
- For a point source the intensity obeys the inverse square law, $I \propto \frac{1}{r^2}$, because the same power spreads over a sphere of area $4\pi r^2$.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Intensity level** | $\beta = 10 \log_{10} \frac{I}{I_0}$ | $\beta$ = intensity level (dB); $I$ = intensity (W m$^{-2}$); $I_0$ = $1.0\times10^{-12}$ W m$^{-2}$ |
| **Intensity from intensity level** | $I = I_0 \times 10^{\beta/10}$ | $I$ = intensity (W m$^{-2}$); $I_0$ = $1.0\times10^{-12}$ W m$^{-2}$; $\beta$ = intensity level (dB) |
| **Difference in intensity level** | $\beta_2 - \beta_1 = 10 \log_{10} \frac{I_2}{I_1}$ | $\beta_1$, $\beta_2$ = intensity levels (dB); $I_1$, $I_2$ = intensities (W m$^{-2}$) |
| **N identical incoherent sources** | $\beta = \beta_1 + 10 \log_{10} N$ | $\beta$ = intensity level of N sources together (dB); $\beta_1$ = intensity level of one source (dB); N = number of identical incoherent sources |
| **Level difference with distance** | $\beta_1 - \beta_2 = 20 \log_{10} \frac{r_2}{r_1}$ | $\beta_1$, $\beta_2$ = intensity levels (dB) at distances $r_1$, $r_2$ (m) from the source; for a point source in free space |
| **Intensity from a point source** | $I = \frac{P}{4\pi r^2}$ | $I$ = intensity (W m$^{-2}$); P = power of the source (W); r = distance from the source (m) |

## TIPS TO ANSWER

- Write the formula first, substitute I and $I_0$ with the same units, evaluate the logarithm, and give $\beta$ in dB to 3 s.f. (or 2 s.f. if the data are 2 s.f.).
- To find I from $\beta$, rearrange: $I = I_0 \times 10^{\beta/10}$; show the power of ten explicitly.
- For several sources, add intensities (W m$^{-2}$) first, then convert; do not add decibels.
- Show the intermediate ratio (e.g. $\frac{I}{I_0} = 2.5\times10^7$) so that the logarithm step is clear.

## SECTION A TRAPS

**Trap:** Adding decibels: two 60 dB sources do not give 120 dB; they give about 63 dB.

**Trap:** Dropping the factor 10 (or using 20) in $\beta = 10 \log \frac{I}{I_0}$; 20 log is for amplitude or pressure ratios only.

**Trap:** Using the natural logarithm ln instead of $\log_{10}$.

**Trap:** Treating 10 dB as doubling the intensity. 10 dB is ×10; ×2 is about 3 dB.

## TRICKS

**Trick:** Memorise: ×2 → +3 dB, ×4 → +6 dB, ×10 → +10 dB, ×100 → +20 dB.

**Trick:** Doubling the distance from a point source lowers the level by 20 log 2 ≈ 6 dB.

# 22.4 Beat

## DEFINITIONS & LAWS

- **Beats** are the periodic variations in intensity (loudness) caused by the superposition of two sound waves of slightly different frequencies.
- Beats are produced because the two waves alternate between being in phase (constructive interference, loud) and in antiphase (destructive interference, soft).
- The **beat frequency** is the number of loud (or soft) sounds heard per second and equals the magnitude of the difference between the two frequencies.
- The frequency of the resultant sound is the mean of the two frequencies, and its amplitude varies at the beat frequency.
- Beats can be used to compare an unknown frequency with a standard; loading or filing one source changes the beat frequency in a predictable way.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Beat frequency** | $f_{beat} = \lvert f_1 - f_2 \rvert$ | $f_{beat}$ = beat frequency (Hz); $f_1$, $f_2$ = frequencies of the two sources (Hz) |
| **Time between loud beats** | $T_{beat} = \frac{1}{f_{beat}}$ | $T_{beat}$ = time between successive loud beats (s); $f_{beat}$ = beat frequency (Hz) |
| **Resultant frequency** | $f_{av} = \frac{f_1 + f_2}{2}$ | $f_{av}$ = resultant frequency (Hz); $f_1$, $f_2$ = frequencies of the two sources (Hz) |
| **Resultant displacement** | $y = 2A\cos(\pi(f_1 - f_2)t)\sin(\pi(f_1 + f_2)t)$ | y = resultant displacement; A = amplitude of each wave; $f_1$, $f_2$ = frequencies (Hz); t = time (s); for equal amplitudes A |

## TIPS TO ANSWER

- Define beats and state $f_{beat} = \lvert f_1 - f_2 \rvert$; for explanation questions, mention the changing phase difference and alternate constructive and destructive interference.
- A frequency unknown from a beat frequency has two possibilities, $f \pm f_{beat}$; give both unless extra information is given.
- For the extra information (filing, waxing or loading a fork), state how the beat frequency changed and deduce which option is eliminated.

- For the time interval, use $T = \frac{1}{f_{beat}}$ with the unit in seconds.

### SECTION A TRAPS

**Trap:** Giving the sum $f_1 + f_2$ or the mean as the beat frequency.

**Trap:** Giving only one value of the unknown frequency. The beat frequency does not show which frequency is higher.

**Trap:** Reversing the effect of filing or loading. Filing raises a fork's frequency; loading with wax lowers it.

**Trap:** Confusing the beat frequency with the interval between beats: 4 Hz means an interval of 0.25 s, not 4 s.

### TRICKS

**Trick:** Filing a fork (f increases): if the beat frequency rises, the fork was already higher than the standard; if it falls, the fork was lower. Opposite for waxing.

**Trick:** Beats need frequencies within a few hertz of each other; a difference of 442 Hz is never a beat frequency.

## 22.5 Doppler Effect

### DEFINITIONS & LAWS

- The **Doppler effect** is the change in the observed frequency of a wave due to relative motion between the source and the observer.
- When a source moves towards a stationary observer, the wavelength is shortened and the observed frequency is higher; when it moves away, the wavelength is lengthened and the frequency is lower.
- When an observer moves towards a stationary source, more wavefronts are intercepted per second, so the observed frequency is higher; moving away gives a lower frequency.
- Speeds are taken relative to the air (medium); the speed of sound in air is unchanged by the motion of source or observer.
- The Doppler shift of waves reflected from a moving object is used in radar speed guns and in medical blood-flow measurement.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Moving source, stationary observer** | $f' = \frac{fv}{v \mp v_s}$ | $f'$ = observed frequency (Hz); $f$ = source frequency (Hz); $v$ = speed of sound (m $s^{-1}$); $v_s$ = speed of source (m $s^{-1}$); upper sign (−) when the source approaches, lower sign (+) when it recedes |
| **Moving observer, stationary source** | $f' = \frac{f(v \pm v_o)}{v}$ | $f'$ = observed frequency (Hz); $f$ = source frequency (Hz); $v$ = speed of sound (m $s^{-1}$); $v_o$ = speed of observer (m $s^{-1}$); plus when the observer approaches, minus when moving away |
| **General Doppler formula** | $f' = \frac{f(v \pm v_o)}{v \mp v_s}$ | $f'$ = observed frequency (Hz); $f$ = source frequency (Hz); $v$ = speed of sound (m $s^{-1}$); $v_o$ = speed of observer (m $s^{-1}$); $v_s$ = speed of source (m $s^{-1}$) |
| **Small-speed approximation** | $\frac{\Delta f}{f} \approx \frac{v_{rel}}{v}$ | $\Delta f$ = change in frequency (Hz); $f$ = frequency (Hz); $v_{rel}$ = relative speed of source and observer (m $s^{-1}$); $v$ = speed of sound (m $s^{-1}$); for small speeds |

- Reflection from a moving surface: apply the formula twice, first with the surface as the observer, then as the source.

## TIPS TO ANSWER

- Identify who is moving, then choose the formula; write the sign choice explicitly (approach: frequency higher).
- Substitute with all speeds in m $s^{-1}$ and compute $f'$ to 3 s.f.; state whether it is higher or lower than $f$.
- Check the sign of the shift against the physical situation before giving the answer.
- For explanation questions, refer to the number of wavefronts reaching the observer per second and the change in wavelength.

## SECTION A TRAPS

**Trap:** Using $v + v_s$ in the denominator for an approaching source. Approach means $v - v_s$, giving a larger $f'$.

**Trap:** Placing the observer's speed in the denominator or the source's in the numerator (swapped roles).

**Trap:** Using the same frequency formula for both a moving source and a moving observer, which are different when speeds are comparable to $v$.

**Trap:** Giving $f'$ lower than $f$ for an approaching source or observer.

## TRICKS

> **Trick:** Direction check: approaching → f′ > f. Eliminate any option on the wrong side of f.

> **Trick:** Source: $v_s$ subtracts from the denominator when approaching. Observer: $v_o$ adds to the numerator when approaching.

Chapter 23

# GEOMETRICAL OPTICS

## 23.1 Spherical Mirrors

### DEFINITIONS & LAWS

- **Sign convention used throughout (real-is-positive):** distances of real objects and real images are positive; virtual images (behind the mirror) have negative v; f and R are positive for a concave mirror and negative for a convex mirror.
- The **centre of curvature** C is the centre of the sphere of which the mirror is a part; the **radius of curvature** r is the radius of that sphere.
- The **pole** P is the centre of the reflecting surface of the mirror; the **aperture** is the size (area) of the reflecting surface.
- The **principal axis** is the line through the pole and the centre of curvature; it is normal to the mirror at its pole.
- The **principal focus** of a concave mirror is the point on the principal axis through which rays parallel and close to the axis converge after reflection.
- The **focal length** is the distance from the pole to the principal focus; $f = \frac{R}{2}$ for a spherical mirror of small aperture, where R is the radius of curvature.
- **Linear magnification** is the ratio of the image height to the object height, $m = \frac{h_i}{h_o} = -\frac{v}{u}$; m is negative for an inverted (real) image and positive for an upright (virtual) image.
- A concave mirror forms a real, inverted image for $u > f$ and a virtual, upright, magnified image for $u < f$; a convex mirror always forms a virtual, upright, diminished image.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Mirror equation** | $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$ | u = object distance; v = image distance; f = focal length (all in m or cm, same unit) |
| **Focal length and radius** | $f = \frac{R}{2}$ | f = focal length; R = radius of curvature; f > 0 for concave, f < 0 for convex |
| **Linear magnification** | $m = -\frac{v}{u} = \frac{h_i}{h_o}$ | m = magnification (no unit); $h_i$ = image height; $h_o$ = object height; $\lvert m\rvert > 1$ magnified, $\lvert m\rvert < 1$ diminished |
| **Image same size as object** | $u = v = 2f = R$; $m = -1$ | u = object distance; v = image distance; f = focal length; R = radius of curvature; real, inverted image |

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Virtual image** | $v < 0$; $m = -\frac{v}{u} > 0$ | v = image distance; u = object distance; m is positive (upright image) |

### TIPS TO ANSWER

- State the convention (real-is-positive) before substituting; write f with the correct sign.
- Find v from $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$, then $m = -\frac{v}{u}$; state the nature: real/virtual, upright/inverted, magnified/diminished.
- Give the position as 'v cm in front of / behind the mirror' in words, in agreement with the sign of v.
- In a ray diagram use at least two principal rays (parallel to axis then through F; through C reflecting back on itself), with arrows and labelled F, C, P.

### SECTION A TRAPS

**Trap:** Using f = R instead of $f = \frac{R}{2}$, or forgetting that R is the distance to the centre of curvature, not the focus.

**Trap:** Taking f of a convex mirror as positive in the real-is-positive convention; it is negative, so v is negative.

**Trap:** Reporting m without sign. The sign tells the orientation: negative = inverted.

**Trap:** Inverting the result: $\frac{1}{v}$ is the reciprocal of v; do not quote $\frac{1}{v}$ as the image distance.

### TRICKS

**Trick:** Limiting case: $u \to \infty$ gives $v = f$; $u = 2f$ gives $v = 2f$ with $m = -1$; $u = f$ gives no image ($v \to \infty$).

**Trick:** If $u < f$ for a concave mirror, $\frac{1}{f} - \frac{1}{u}$ is negative, so the image is virtual and magnified.

## 23.2 Refraction at Spherical Surfaces

### DEFINITIONS & LAWS

- **Sign convention (real-is-positive):** u is positive for a real object in front of the surface; v is positive for a real image formed on the side where light emerges, negative for a virtual image on the object side; R is positive when the centre of curvature lies on the outgoing (emerging) side, i.e. a surface convex towards the incident light.
- Refraction at a spherical surface obeys Snell's law $n_1 \sin i = n_2 \sin r$ at each point; for paraxial rays this leads to the surface equation below.
- **Apparent depth**: an object in a denser medium appears closer to the surface when viewed normally, because rays bend away from the normal on leaving it.
- A plane surface has $R \to \infty$, so the term $\frac{n_2 - n_1}{R}$ vanishes; then $\frac{n_1}{u} + \frac{n_2}{v} = 0$ and $\frac{v}{u} = -\frac{n_2}{n_1}$.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Refraction at a spherical surface** | $\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R}$ | $n_1$ = refractive index of the medium containing the object; $n_2$ = refractive index of the other medium; u = object distance; v = image distance; R = radius of curvature (m) |
| **Apparent depth** | Apparent depth = real depth × $\frac{\text{n of observer's medium}}{\text{n of object's medium}}$; water viewed from air: apparent depth = real depth/n | Depths in m; n = refractive index of water for the water-from-air case |
| **Second focal length** | $f_2 = \frac{n_2 R}{n_2 - n_1}$ | $f_2$ = second focal length (m), obtained from $u \to \infty$; $n_1$, $n_2$ = refractive indices; R = radius of curvature (m) |
| **First focal length** | $f_1 = \frac{n_1 R}{n_2 - n_1}$ | $f_1$ = first focal length (m), obtained from $v \to \infty$; $n_1$, $n_2$ = refractive indices; R = radius of curvature (m) |
| **Lateral magnification** | $m = -\frac{n_1 v}{n_2 u}$ | m = lateral magnification (no unit); $n_1$, $n_2$ = refractive indices; u = object distance; v = image distance |

## TIPS TO ANSWER

- State which medium is $n_1$ (object side) and which is $n_2$, then give the sign of R with a reason (centre on outgoing side: positive).
- Substitute with consistent units, solve for $\frac{1}{v}$ and then v; the sign of v shows whether the image is real (outgoing side) or virtual (object side).
- Quote the position relative to the surface: 'v cm inside the glass' or 'v cm on the object side'.
- For plane surfaces, use the apparent depth relation and give the answer in the same unit as the real depth.

### SECTION A TRAPS

**Trap:** Interchanging $n_1$ and $n_2$, so that the numerator $(n_2 - n_1)$ changes sign and v is wrong.

**Trap:** Using the wrong sign of R for a concave surface (R negative when the centre of curvature is on the incident side).

**Trap:** Applying apparent depth = real depth × n. It is divided by n when looking from air into the denser medium.

**Trap:** Treating the refracting surface as a thin lens; the surface equation involves two indices and a single radius.

### TRICKS

**Trick:** Check the sign of v by physical reasoning: from a less dense medium into a denser one through a convex surface, the rays converge more, so v is positive and shorter than in the plane case.

**Trick:** Plane surface test: $v = -\frac{n_2}{n_1}u$, so for a fish in water viewed from air v is shorter than u.

## 23.3 Thin Lenses

### DEFINITIONS & LAWS

- **Sign convention (real-is-positive):** u positive for a real object; v positive for a real image (on the opposite side of the lens), negative for a virtual image; f positive for a converging (convex) lens and negative for a diverging (concave) lens.
- The **principal focus** of a converging lens is the point on the axis to which rays parallel and close to the axis converge after refraction through the lens.
- The **power** of a lens is the reciprocal of its focal length in metres, $P = \frac{1}{f}$, with unit dioptre (D).
- **Linear magnification** $m = \frac{h_i}{h_o} = -\frac{v}{u}$; a real image from a single lens is inverted (m negative); a virtual image is upright (m positive).
- A convex lens forms a real, inverted image when $u > f$ and a virtual, upright, magnified image when $u < f$ (magnifying glass); a concave lens always forms a virtual, upright, diminished image.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Thin lens equation** | $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$ | u = object distance; v = image distance; f = focal length (all distances in the same unit) |
| **Lensmaker's formula** | $\frac{1}{f} = (n - 1)(\frac{1}{R_1} + \frac{1}{R_2})$ | n = refractive index of lens material; $R_1$, $R_2$ = radii of the two surfaces, each positive when its centre of curvature lies on the side from which the light emerges (biconvex: both positive); if the form $\frac{1}{R_1} - \frac{1}{R_2}$ is used, $R_2$ is negative for a biconvex lens |
| **Linear magnification** | $m = -\frac{v}{u}$ | m = magnification (no unit); u = object distance; v = image distance |
| **Power of a lens** | $P = \frac{1}{f}$ | P = power (dioptre, D); f = focal length (m) |
| **Lenses in contact** | $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$; $P = P_1 + P_2$ | f = combined focal length; $f_1$, $f_2$ = focal lengths of the lenses; P = combined power (D); $P_1$, $P_2$ = powers of the lenses (D) |
| **Real image same size as object** | $u = v = 2f$ | u = object distance; v = image distance; f = focal length |
| **Minimum object-image distance** | $u + v = 4f$ | Minimum distance between object and a real image is 4f; f = focal length |

## TIPS TO ANSWER

- State the sign convention, write f with the correct sign, and find v from $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$; then give m and describe the image (real/virtual, inverted/upright, magnified/diminished).
- For combinations in contact, add the powers or reciprocal focal lengths and show the working to 3 s.f.
- Convert f to metres before calculating power; give the unit D.
- For ray diagrams, use two principal rays (parallel to axis then through F; through the optical centre undeviated), arrows, and label F and the optical centre.

## SECTION A TRAPS

**Trap:** Calculating power with f in cm (a 25 cm lens is 4.0 D, not 0.040 D).

**Trap:** Adding focal lengths instead of reciprocals for two lenses in contact: the combination of 25 cm and 40 cm is 15.4 cm, not 65 cm.

**Trap:** Using the lensmaker's formula with the sign convention for one radius reversed; check that a biconvex lens in air gives a positive f.

**Trap:** Quoting the sign of v the wrong way round: in real-is-positive a negative v is a virtual image on the object side.

## TRICKS

**Trick:** Unit check: P in D requires f in m; ×100 slip is the commonest error.

**Trick:** Magnification ratio method: a real image m times larger has v = mu, so $\frac{1}{f} = (1 + \frac{1}{m})/u$ and $u = f(1 + \frac{1}{m})$.

Chapter 24

# WAVE OPTICS

## 24.1 Huygens's Principle

### DEFINITIONS & LAWS

- **Wavefront** is a line or surface joining all points of a wave that are in phase (same displacement and same direction of motion).
- **Huygens's principle**: every point on a wavefront acts as a source of secondary spherical wavelets that spread out with the speed of the wave; the new wavefront after a short time t is the envelope (common tangent) of these wavelets.
- **Diffraction** is the spreading of a wave into the geometrical shadow when it passes the edge of an obstacle or through a gap; the points at the edge act as sources of secondary wavelets that spread into the shadow region.
- **Reflection and refraction**: construction with secondary wavelets gives the law of reflection (angle of incidence = angle of reflection) and Snell's law; on refraction the frequency is unchanged while the speed and wavelength change.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Wave speed** | $v = f\lambda$ | $v$ = wave speed (m $s^{-1}$); $f$ = frequency (Hz); $\lambda$ = wavelength (m) |
| **Refractive index** | $n = \frac{c}{v} = \frac{\lambda_0}{\lambda}$ | $n$ = refractive index of the medium (no unit); $c$ = speed in vacuum (m $s^{-1}$); $v$ = speed in the medium (m $s^{-1}$); $\lambda_0$ = wavelength in vacuum (air) (m); $\lambda$ = wavelength in the medium (m) |
| **Snell's law** | $n_1 \sin\theta_1 = n_2 \sin\theta_2$ | $n_1$, $n_2$ = refractive indices of the two media; $\theta_1$, $\theta_2$ = angles measured from the normal |
| **Wavelet radius** | $r = vt$ | Distance moved by a wavefront in time t, which is the radius of each secondary wavelet; $v$ = wave speed (m $s^{-1}$); $t$ = time (s) |

### TIPS TO ANSWER

- For a Huygens construction, draw several points on the wavefront, draw arcs of radius vt, then draw the common tangent as the new wavefront.
- State that the secondary wavelets have the same speed and frequency as the original wave.

- For diffraction, state that the edge points act as sources and the wavelets spread into the region behind the obstacle.
- For wavelength in a medium: state $\lambda = \frac{\lambda_0}{n}$, f unchanged, then substitute with units.

### SECTION A TRAPS

**Trap:** The frequency of light does not change when it enters glass or water. Options that say the frequency is multiplied or divided by n are wrong; only the wavelength and the speed change.

**Trap:** Plane wavefronts are not 'perfectly flat everywhere'. A small part of a spherical wavefront from a distant source is approximately plane; the radius of curvature is very large, not zero.

**Trap:** Wavefronts are drawn perpendicular to rays. Mixing up a ray (direction of travel) with a wavefront (surface of constant phase) loses the mark.

### TRICKS

**Trick:** Wavelength in a medium of refractive index n: divide by n (shorter), e.g. 600 nm in glass of n = 1.50 gives 400 nm. Never multiply.

**Trick:** If an option says the speed, frequency and wavelength all change on entering a new medium, eliminate it: f is fixed by the source.

## 24.2 Interference

### DEFINITIONS & LAWS

- **Principle of superposition**: when two or more waves meet at a point, the resultant displacement is the vector sum of the displacements of the individual waves at that point.
- **Interference** is the superposition of two or more coherent waves, producing a pattern of reinforcement (maxima) and cancellation (minima).
- **Coherent sources** are sources that emit waves of the same frequency (wavelength) with a constant phase difference.
- **Optical path** between two points in a medium is the distance in vacuum that contains the same number of wavelengths as the medium between the two points, for light of the same frequency; optical path = nd, where n is the refractive index and d the actual distance.
- **Constructive interference** occurs when the waves meet in phase (path difference = nλ, n = 0, 1, 2, ...); **destructive interference** occurs when they meet in antiphase (path difference = (n + ½)λ).
- Two separate light bulbs do not give a stable interference pattern because the phase relationship of their light changes randomly and rapidly, so they are not coherent.

- On reflection from a medium of higher refractive index (denser medium) the wave undergoes a phase change of π (equivalent to a path change of $\frac{\lambda}{2}$); there is no phase change on reflection at a less dense medium.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Constructive interference** | $\Delta = n\lambda$ | $\Delta$ = path difference (m); $\lambda$ = wavelength (m); n = 0, 1, 2, ... |
| **Destructive interference** | $\Delta = (n + ½)\lambda$ | $\Delta$ = path difference (m); $\lambda$ = wavelength (m); n = 0, 1, 2, ... |
| **Phase difference** | $\varphi = \frac{2\pi}{\lambda} \times \Delta$ | $\varphi$ = phase difference (rad); $\Delta$ = path difference (m); $\lambda$ = wavelength (m) |
| **Intensity and amplitude** | $I \propto A^2$ | I = intensity (W $m^{-2}$); A = amplitude |
| **Maximum and minimum intensity** | $I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$; $I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$ | Two coherent waves of intensities $I_1$ and $I_2$ (W $m^{-2}$) |
| **Maximum and minimum amplitude** | $A_{max} = A_1 + A_2$; $A_{min} = \lvert A_1 - A_2 \rvert$ | Amplitudes add; $A_1$, $A_2$ = amplitudes of the two waves |

## TIPS TO ANSWER

- For path difference questions: divide $\Delta$ by $\lambda$. A whole number gives a bright (maximum); a half-integer gives a dark (minimum). State the conclusion with the value of $\frac{\Delta}{\lambda}$.
- For 'explain why no pattern' state: the sources are not coherent, i.e. the phase difference is not constant.
- For an intensity ratio: add and subtract the square roots of the intensities (amplitudes), then square.
- Mention any phase change of π on reflection before deciding between maximum and minimum.

## SECTION A TRAPS

**Trap:** Half-integer path difference is destructive. A path difference of 2.5λ is NOT 'close to a whole number, so constructive'; a path difference of 2.0λ or 3.0λ is bright.

**Trap:** Intensities do not simply add or subtract at the extremes. With $I_1 = 4I_0$ and $I_2 = I_0$: $\frac{I_{max}}{I_{min}} = \frac{(2 + 1)^2}{(2 - 1)^2} = 9$, not 3 (forgetting to square), not $\frac{5}{3}$ (which is $\frac{I_1 + I_2}{I_1 - I_2}$) and not 4.

**Trap:** Interference does not need waves of different types, and two lamps of the same colour are still incoherent. Coherence, not colour, is the requirement.

### TRICKS

**Trick:** Convert the path difference to a multiple of λ first $\frac{\Delta}{\lambda}$ and decide by inspection: integer = bright, half-integer = dark.

**Trick:** For $\frac{I_{max}}{I_{min}}$ with amplitude ratio $r = \frac{A_1}{A_2}$: ratio = $(\frac{r+1}{r-1})^2$. Intensity ratio 4 : 1 means r = 2, so the ratio is 9.

## 24.3 Two-slit Interference Pattern

### DEFINITIONS & LAWS

- **Young's double-slit experiment**: light from a single narrow slit (or a laser) illuminates two narrow slits $S_1$ and $S_2$ which act as coherent sources; the overlapping diffracted waves produce alternate bright and dark fringes on a screen.
- The bright fringe is formed where the path difference $S_2P - S_1P = n\lambda$; the dark fringe where $S_2P - S_1P = (n + ½)\lambda$.
- Fringes are equally spaced and of nearly equal brightness near the centre; the central fringe (n = 0) is bright because the path difference is zero.
- With white light, the central fringe is white and the other fringes are coloured (violet nearest the centre, red furthest) since fringe spacing $\propto \lambda$.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Fringe spacing** | $x = \frac{\lambda D}{a}$ | x = fringe spacing (m); λ = wavelength (m); D = slit-to-screen distance (m); a = slit separation (m); valid for $D \gg a$ |
| **Path difference at a point** | $\Delta = \frac{ay}{D}$ | Δ = path difference (m); y = distance from the central fringe (m) |
| **Bright fringe position** | $y_n = \frac{n\lambda D}{a}$ | $y_n$ = position of the nth bright fringe (m); n = 0, 1, 2, … |
| **Dark fringe position** | $y_n = \frac{(n + ½)\lambda D}{a}$ | $y_n$ = position of the nth dark fringe (m) |

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Angular position of bright fringe** | $a \sin \theta = n\lambda$ | $\theta$ = angular position of the nth bright fringe; a = slit separation (m); $\lambda$ = wavelength (m) |
| **Wavelength and spacing in a medium** | $\lambda' = \frac{\lambda}{n}$; $x' = \frac{x}{n}$ | In a medium of refractive index n; $\lambda'$ = wavelength in the medium (m); x′ = fringe spacing in the medium (m) |

## TIPS TO ANSWER

- Start from $x = \frac{\lambda D}{a}$, rearrange for the unknown, convert mm and nm to m, substitute with units, give 3 s.f.
- For 'distance between the mth and the nth fringes' use (n − m)x, not n x or m x separately.
- For effects of changing a quantity, state the proportionality: $x \propto \lambda$, $x \propto D$, $x \propto \frac{1}{a}$, then the new value.
- State assumptions: slits narrow, $a \ll D$ so that $\sin\theta \approx \tan\theta \approx \theta$.

## SECTION A TRAPS

**Trap:** Slit separation a is in the denominator: halving a doubles x. Options that give 'half' for halved a are wrong.

**Trap:** Unconverted units: 0.30 mm = $0.30 \times 10^{-3}$ m and 600 nm = $600 \times 10^{-9}$ m. A wrong power of ten is the most common slip.

**Trap:** Immersing the apparatus in a liquid DEcreases the fringe spacing ($x' = \frac{x}{n}$), it does not increase it. For n = 1.33, 1.60 mm becomes 1.20 mm, not 2.13 mm.

**Trap:** The angular position of the 3rd bright fringe uses m = 3, not m = 1; and with small angles the answer in degrees is very small (0.215°), not 3°.

## TRICKS

**Trick:** Ratio method: if only one quantity changes, $x_2 = x_1 \times$ (factor), no need to compute the full expression.

**Trick:** Check the answer is sensible: for visible light, a ~ 0.1–1 mm and D ~ 1–2 m, x is of the order of mm.

# 24.4 Interference in A Thin Film

## DEFINITIONS & LAWS

- **Thin-film interference** is the interference between the light reflected from the top surface and the light reflected from the bottom surface of the film; the two beams are coherent because they originate from the same incident wave.
- A phase change of π occurs on reflection when the light is reflected from a medium of higher refractive index; there is no phase change on reflection from a medium of lower refractive index.
- **Colours of a thin film in white light**: the film thickness varies, so different wavelengths satisfy the condition for constructive interference at different places, giving coloured bands.
- **Air wedge**: the air layer between two glass plates in contact at one edge; the fringes are straight, parallel and equally spaced, and the fringe at the line of contact is dark because of the π phase change at one reflection only.
- **Non-reflective coating**: a thin film of refractive index between those of air and glass; the reflected beams from the two surfaces interfere destructively so that reflection is reduced.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Thin film, one phase change: constructive** | $2nt = (m + ½)\lambda$ | Film of index n in air, one phase change at the top surface; t = film thickness (m); λ = wavelength in air (m); m = 0, 1, 2, ... |
| **Thin film, one phase change: destructive** | $2nt = m\lambda$ | Film of index n in air, one phase change at the top surface; m = 0, 1, 2, ... |
| **Thin film, no relative phase change: constructive** | $2nt = m\lambda$ | Both reflections with a π change, or neither (e.g. coating on glass with $n_{air} < n_{film} < n_{glass}$) |
| **Thin film, no relative phase change: destructive** | $2nt = (m + ½)\lambda$ | Both reflections with a π change, or neither (e.g. coating on glass with $n_{air} < n_{film} < n_{glass}$) |
| **Minimum coating thickness** | $t = \frac{\lambda}{4n}$ | Minimum coating thickness for destructive reflection of wavelength λ (m); n = refractive index of the coating |
| **Air wedge dark fringes** | $2t = m\lambda$ | Air wedge at normal incidence; dark fringes; t = air gap thickness (m); m = 0, 1, 2, ... |

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Air wedge fringe spacing** | $x = \frac{\lambda}{2\theta}$ | Air wedge at normal incidence; x = fringe spacing (m); θ = wedge angle (rad) |

## TIPS TO ANSWER

- First decide the number of π phase changes at the two surfaces (list $n_1$, $n_2$, $n_3$ in order), then write the correct condition.
- State that the optical path difference is 2nt (the beam travels t down and t up inside the film) and that λ in the formula is the wavelength in air (the n is already included).
- Rearrange for t or λ, substitute, give the answer with unit, and name the order m used.
- For 'explain the colours', mention variation in thickness and the condition for each wavelength.

## SECTION A TRAPS

**Trap:** Check BOTH surfaces. For a soap film in air only the top surface gives the π change; for a coating (air → $MgF_2$ → glass) both surfaces do, so the conditions swap and $t = \frac{\lambda}{4n}$ gives destructive reflection.

**Trap:** Using $\frac{\lambda}{2n}$ instead of $\frac{\lambda}{4n}$: the minimum thickness for the non-reflective condition $2nt = \frac{\lambda}{2}$ is $t = \frac{\lambda}{4n}$, not $\frac{\lambda}{2n}$ or $\frac{\lambda}{n}$.

**Trap:** The wavelength in the film is $\frac{\lambda}{n}$. Do not use λ in the film for the path difference in addition to the factor n in 2nt (double counting).

## TRICKS

**Trick:** A film that is very thin ($t \to 0$) compared with the wavelength appears dark in reflection if there is one π change, so the first bright condition is at $t = \frac{\lambda}{4n}$.

**Trick:** Quick check on a soap-film answer: 560 nm light, n = 1.33 gives $t_{min} = \frac{560}{4 \times 1.33} = 105$ nm for strong reflection.

# 24.5 Diffraction by A Single Slit

## DEFINITIONS & LAWS

- **Diffraction** is the spreading of waves when they pass through a gap or round an obstacle; it is most marked when the width of the gap is comparable with the wavelength.
- A single-slit diffraction pattern has a wide, bright central maximum with weaker secondary maxima on both sides; the intensity of the secondary maxima falls rapidly.
- The pattern arises from interference between the secondary wavelets from different points across the width of the slit.
- The central maximum is twice as wide as each secondary maximum; the minima occur where the path difference between the two edges of the slit is a whole number of wavelengths.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Single-slit minima** | $a \sin \theta = m\lambda$ | m = 1, 2, 3, ... for the minima; a = slit width (m); θ = angle from the central axis (rad or °); λ = wavelength (m) |
| **Angular width of central maximum** | $2\theta_1$; $\sin \theta_1 = \frac{\lambda}{a}$ | $2\theta_1$ = angular width of the central maximum (rad); $\theta_1$ = angle of the first minimum; a = slit width (m); λ = wavelength (m) |
| **Width of central maximum on screen** | $w = \frac{2\lambda D}{a}$ | small angles; w = width of the central maximum (m); D = distance from slit to screen (m); λ = wavelength (m); a = slit width (m) |
| **Small-angle approximation** | $\theta \approx \frac{\lambda}{a}$ | for small angles; θ = angle of the first minimum (rad); λ = wavelength (m); a = slit width (m) |

## TIPS TO ANSWER

- Write the condition for minima first ($a \sin \theta = m\lambda$), then substitute with values converted to metres.
- For the width of the central maximum, double the distance from the centre to the first minimum: $w = \frac{2\lambda D}{a}$. State clearly 'width' versus 'half-width'.
- For 'what happens when a decreases', state: sin θ increases so the central maximum becomes wider (and less intense).
- A sketch of intensity against angle needs a labelled axis, a tall central peak that is twice the width of the others and small secondary peaks.

## SECTION A TRAPS

**Trap:** A narrower slit gives a WIDER pattern ($w \propto \frac{1}{a}$). Options saying the pattern becomes narrower or disappears are wrong.

**Trap:** Mixing up the angles: the first minimum is at $a \sin\theta = \lambda$, not at $a \sin\theta = \frac{\lambda}{2}$. The condition $a \sin\theta = m\lambda$ is for minima, not for maxima.

**Trap:** Forgetting the factor 2: the angular width of the central maximum is $2\theta_1$ (for example $2 \times 2.87° = 5.7°$), not $\theta_1$.

**Trap:** Do not confuse a (slit width) with d (slit separation or grating spacing) in the formula.

### TRICKS

**Trick:** If $a \gg \lambda$ the pattern is hardly spread; if $a < \lambda$ no minimum exists ($\sin\theta$ cannot exceed 1) and the light spreads over a wide region.

**Trick:** Width ratio: $\frac{w_2}{w_1} = \frac{\lambda_2}{\lambda_1}\frac{a_1}{a_2}\frac{D_2}{D_1}$ — use the proportionality instead of recalculating.

## 24.6 Diffraction Gratings

### DEFINITIONS & LAWS

- **Diffraction grating** is an optical device with a large number of equally spaced parallel slits (lines) of very narrow width; the diffracted beams interfere to produce sharp, bright maxima at definite angles.
- **Grating spacing d** is the distance between the centres of adjacent slits; $d = \frac{1}{N}$, where N is the number of lines per unit length.
- The maxima occur where the path difference between light from adjacent slits is a whole number of wavelengths.
- With white light, the zero-order maximum is white and each higher order is a spectrum with violet nearest and red furthest from the centre; the higher orders may overlap.
- Increasing the number of lines illuminated makes the maxima sharper and brighter; the angular positions of the maxima depend on d and $\lambda$ only.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Grating equation** | $d \sin \theta = m\lambda$ | m = 0, 1, 2, ... (order); d = grating spacing (m); θ = angle of the mth order from the normal (rad or °); λ = wavelength (m) |
| **Grating spacing** | $d = \frac{1}{N}$ | N = number of lines per metre ($m^{-1}$); d = grating spacing (m); for example 500 lines $mm^{-1}$ = 5.00 × $10^5$ lines $m^{-1}$, d = 2.00 × $10^{-6}$ m |
| **Highest order** | $m_{max} < \frac{d}{\lambda}$ | $m_{max}$ = largest integer less than $\frac{d}{\lambda}$ (because sin θ < 1); d = grating spacing (m); λ = wavelength (m) |
| **Total number of maxima** | $2m_{max} + 1$ | total number of maxima observed; $m_{max}$ = highest order |
| **Grating angle from screen** | $\tan \theta = \frac{y}{D}$ | y = distance on the screen from the central maximum (m); D = distance from grating to screen (m); then find sin θ from θ |

## TIPS TO ANSWER

- Calculate d from the number of lines per mm or cm FIRST, in metres, then apply $d \sin \theta = m\lambda$.
- For the maximum order, evaluate $\frac{d}{\lambda}$ and take the whole number part; state that sin θ cannot exceed 1.
- When the distance y on the screen is given, find $\theta = \tan^{-1}\frac{y}{D}$, then use sin θ in the grating equation (do not use $\frac{y}{D}$ directly for sin θ unless the angle is small).
- State the answer to 3 s.f. with the order m used.

## SECTION A TRAPS

**Trap:** Calculating d the wrong way round: for 600 lines per mm, $d = \frac{1}{6.00 \times 10^5} = 1.67 \times 10^{-6}$ m, not $600 \times 10^{-3}$ m. Remember d is the reciprocal of N.

**Trap:** Taking the highest order as $\frac{d}{\lambda}$ rounded up. For $\frac{d}{\lambda} = 3.40$ the highest order is 3, not 4; for 2.78 it is 2, not 3.

**Trap:** Using the 2nd order angle as twice the 1st order angle: it is the sine that is proportional to m, not the angle. For $d \sin \theta_1 = \lambda$ and $\sin \theta_2 = 2 \sin \theta_1$.

## TRICKS

**Trick:** Check sin θ ≤ 1 at once: if $\frac{m\lambda}{d} > 1$, that order does not exist.

**Trick:** More lines per mm means smaller d, therefore larger angles and fewer orders; use this to eliminate options quickly.

# 24.7 Polarisation

## DEFINITIONS & LAWS

- **Plane-polarised (linearly polarised) light** is light in which the electric field vibrations are in one plane only, perpendicular to the direction of propagation.
- **Unpolarised light** is light whose electric field vibrations occur in all planes perpendicular to the direction of propagation; it can be resolved into two mutually perpendicular components of equal intensity, both perpendicular to the direction of propagation.
- **Plane of polarisation** is the plane containing the electric field vibrations of the wave and the direction of propagation.
- Only transverse waves can be polarised; the polarisation of light shows that light is a transverse wave, whereas sound (a longitudinal wave) cannot be polarised.
- **Polariser** produces plane-polarised light from unpolarised light; **analyser** is a second polariser used to detect the plane of polarisation.
- **Malus's law**: the intensity of plane-polarised light transmitted through an analyser is proportional to the square of the cosine of the angle between the plane of polarisation of the light and the transmission axis of the analyser.
- **Brewster's law**: when light is incident at the polarising angle $\theta_B$, the reflected light is completely plane-polarised (vibrations perpendicular to the plane of incidence) and the reflected and refracted rays are perpendicular to each other.
- Unpolarised light passing through an ideal polariser is reduced to half of the incident intensity.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Malus's law** | $I = I_0 \cos^2 \theta$ | $I_0$ = intensity of plane-polarised light incident on the analyser (W m$^{-2}$); I = transmitted intensity (W m$^{-2}$); θ = angle between the plane of polarisation and the transmission axis |
| **Unpolarised light through a polariser** | $I_1 = \frac{I_{unpolarised}}{2}$ | $I_1$ = intensity after the first polariser (W m$^{-2}$); $I_{unpolarised}$ = incident intensity (W m$^{-2}$); then apply Malus's law for each further polariser |
| **Brewster's law** | $\tan \theta_B = \frac{n_2}{n_1}$; $\tan \theta_B = n$ | $\theta_B$ = Brewster (polarising) angle; $n_1$, $n_2$ = refractive indices of the first and second media; $\tan \theta_B = n$ for light incident from air |

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Reflected and refracted rays at Brewster angle** | $\theta_B + \theta_r = 90°$ | $\theta_B$ = Brewster angle; $\theta_r$ = angle of refraction |

## TIPS TO ANSWER

- For unpolarised light: halve the intensity at the first polariser, then apply $I = I_0 \cos^2 \theta$ at each later polariser, using the angle between successive axes.
- For three polarisers, multiply the $\cos^2$ factors of successive pairs of axes (do not use the angle to the first axis for the third).
- For Brewster: state $\tan \theta_B = n$, substitute, and give $\theta_B$ in degrees; mention that the reflected ray is completely polarised.
- Name the type of wave: only transverse waves can be polarised.

## SECTION A TRAPS

**Trap:** Applying Malus's law to unpolarised light directly. The first polariser gives $\frac{I_0}{2}$ (average of $\cos^2$ is ½), and only then does $\cos^2 \theta$ apply: 100 W m$^{-2}$ → 50 W m$^{-2}$ → 37.5 W m$^{-2}$ at 30°.

**Trap:** Using the angle to the FIRST polariser for the third polariser. For crossed 1st and 3rd axes with a 45° polariser between them, each step is $\cos^2 45° = ½$, so the final intensity is $\frac{I_0}{2} \times ½ \times ½ = \frac{I_0}{8}$ (12.5%), not 0.

**Trap:** $\tan \theta_B = n$, not $\sin \theta_B = n$. Water n = 1.33 gives 53.1°; the complementary angle 36.9° is the refraction angle, not the Brewster angle.

**Trap:** Using $\cos \theta$ instead of $\cos^2 \theta$: at 60° the transmitted intensity is $\frac{I_0}{4}$, not $\frac{I_0}{2}$.

## TRICKS

**Trick:** Remember $\cos^2$ values: 30° gives $\frac{3}{4}$, 45° gives $\frac{1}{2}$, 60° gives $\frac{1}{4}$.

**Trick:** Inserting a polariser at 45° between crossed polarisers lets light through (a classic result); two crossed polarisers alone transmit zero.

# 24.8 Optical Waveguides

## DEFINITIONS & LAWS

- **Optical fibre** is a thin strand of glass or plastic consisting of a core surrounded by a cladding of lower refractive index; light is guided along the core by total internal reflection.
- **Total internal reflection** occurs when light travels from a denser to a less dense medium and the angle of incidence exceeds the critical angle.
- **Critical angle $\theta_c$** is the angle of incidence in the denser medium for which the angle of refraction in the less dense medium is 90°.
- Conditions for guiding: $n_{core} > n_{cladding}$ and the angle of incidence at the core–cladding boundary greater than $\theta_c$.
- **Attenuation** is the loss of power as the signal travels along the fibre, caused by absorption and scattering in the glass.
- **Modal dispersion** is the spreading (broadening) of a pulse because rays following different paths take different times; it is reduced in a very narrow core (single-mode) fibre or a graded-index fibre.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Critical angle** | $\sin\theta_c = \frac{n_2}{n_1}$ | $\theta_c$ = critical angle; $n_1$ = refractive index of the core; $n_2$ = refractive index of the cladding |
| **Numerical aperture** | $NA = \sin\theta_a = \sqrt{n_1^2 - n_2^2}$ | $\theta_a$ = maximum acceptance angle (from air); NA = numerical aperture (no unit); $n_1$ = refractive index of the core; $n_2$ = refractive index of the cladding |
| **Attenuation** | $\text{Attenuation} = \frac{10}{L}\log_{10}\frac{P_{in}}{P_{out}}$ | in dB per unit length; L = length of the fibre; $P_{in}$ = input power (W); $P_{out}$ = output power (W) |
| **Pulse transit time along axis** | $t = \frac{L}{v} = \frac{nL}{c}$ | t = time for a pulse along the axis (s); L = length of the fibre (m); $v = \frac{c}{n}$ = speed of light in the core (m $s^{-1}$); n = refractive index of the core; c = speed of light in vacuum (m $s^{-1}$) |

## TIPS TO ANSWER

- Identify $n_1$ (core, larger) and $n_2$ (cladding, smaller) before using $\sin\theta_c = \frac{n_2}{n_1}$; the ratio must be less than 1.
- For NA: substitute the two indices, give 3 s.f.; a numerical aperture is dimensionless.
- For an explanation of guiding, state both conditions: core index greater than cladding index AND angle of incidence at the boundary greater than the critical angle.
- For a time along the fibre, use $v = \frac{c}{n}$, not $v = c$.

## SECTION A TRAPS

**Trap:** Inverting the ratio: sin $\theta_c = \frac{n_2}{n_1}$ (small over large). For core 1.50 and cladding 1.45, $\theta_c$ = 75.2°, not 43.6°.

**Trap:** Saying the angle must be SMALLER than the critical angle for total internal reflection; it must be GREATER than $\theta_c$ (measured from the normal).

**Trap:** Forgetting that the core has the higher index; a core with lower index than the cladding cannot guide light.

**Trap:** Using the speed of light c for the transit time in the glass core: $t = \frac{nL}{c}$; for 2.0 km and n = 1.50, t = 10 µs, not 6.7 µs.

## TRICKS

**Trick:** Critical angle is large when the two indices are close; glass–air (n = 1.50) gives 41.8°.

**Trick:** NA = $\sqrt{(n_1 + n_2)(n_1 - n_2)}$ is quicker than squaring: $\sqrt{2.95 \times 0.05}$ = 0.384.

Chapter 25

# QUANTUM PHYSICS

## 25.1 Photon

### DEFINITIONS & LAWS

- **Photon** is a quantum (packet) of electromagnetic radiation of energy E = hf; it is a neutral particle with zero rest mass that travels at the speed of light.
- **Photoelectric effect** is the emission of electrons from a metal surface when electromagnetic radiation of sufficiently high frequency falls on it.
- **Compton effect** is the scattering of X-ray photons by electrons in atoms; the scattered photon has a longer wavelength (lower energy) than the incident photon, showing that a photon carries momentum.
- **Work function φ** is the minimum energy required to remove an electron from the surface of a metal; **threshold frequency $f_0$** is the minimum frequency for emission, $f_0 = \frac{\varphi}{h}$.
- **Stopping potential $V_s$** is the minimum reverse potential difference that reduces the photocurrent to zero; $eV_s = K_{max}$.
- **Einstein's photoelectric equation**: $hf = \varphi + K_{max}$, an electron absorbs one photon whose energy is shared between the work function and the kinetic energy.
- Observations that wave theory cannot explain: a threshold frequency exists; $K_{max}$ is independent of intensity and depends on frequency; emission is instantaneous.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Photon energy** | $E = hf = \frac{hc}{\lambda}$ | E = photon energy (J); f = frequency (Hz); λ = wavelength (m); h = 6.63 × $10^{-34}$ J s; c = 3.00 × $10^{8}$ m $s^{-1}$ |
| **Photoelectric equation** | $hf = \varphi + ½mv^2_max$; $\varphi = hf_0$ | f = frequency of incident radiation (Hz); φ = work function (J); $f_0$ = threshold frequency (Hz); m = electron mass (kg); $v_{max}$ = maximum speed of photoelectrons (m $s^{-1}$) |
| **Stopping potential** | $eV_s = ½mv^2_max = hf - \varphi$ | $V_s$ = stopping potential (V); e = electron charge (C); m = electron mass (kg); $v_{max}$ = maximum speed of photoelectrons (m $s^{-1}$); φ = work function (J) |
| **Photon momentum** | $p = \frac{h}{\lambda}$ | p = photon momentum (kg m $s^{-1}$); h = Planck constant (J s); λ = wavelength (m) |

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Electronvolt** | $1\ \text{eV} = 1.60 \times 10^{-19}\ \text{J}$; $E\ (\text{eV}) \approx \frac{1240}{\lambda}\ (\text{nm})$ | E (eV) = photon energy in electronvolts; λ (nm) = wavelength in nanometres |

- Graph of $K_{max}$ against f: straight line, gradient = h, intercept on the f-axis = $f_0$, intercept on the energy axis = $-\varphi$.

## TIPS TO ANSWER

- Write $E = hf$ or $\frac{hc}{\lambda}$, convert nm to m and eV to J (or J to eV) consistently, then substitute with units.
- For photoelectric questions write $hf = \varphi + K_{max}$, then $K_{max} = hf - \varphi$; convert both energies to the same unit before subtracting.
- For explanations: state that intensity ∝ number of photons per second ∝ number of photoelectrons per second, and $K_{max}$ depends on frequency only.
- Say 'below the threshold frequency no emission occurs, no matter how intense the light'.

## SECTION A TRAPS

**Trap:** Increasing intensity increases the number of photoelectrons (the saturation current), not $K_{max}$ and not the stopping potential.

**Trap:** Subtracting energies in different units: $hf = 4.64 \times 10^{-19}$ J = 2.90 eV; $\varphi$ = 2.3 eV gives $K_{max}$ = 0.60 eV. Giving 2.90 eV (forgetting $\varphi$) or 5.20 eV (adding $\varphi$) are the usual wrong options.

**Trap:** Threshold wavelength $\lambda_0 = \frac{hc}{\varphi}$ is the LONGEST wavelength that causes emission; shorter wavelengths emit electrons. Do not reverse the inequality.

**Trap:** Reporting hf (in joules) when the options are in electron-volts, or stating the stopping potential in eV (it is in volts: $V_s = \frac{K_{max}}{e}$).

## TRICKS

**Trick:** Use $E\ (\text{eV}) \approx \frac{1240}{\lambda}\ (\text{nm})$ for fast checks: 632.8 nm gives about 1.96 eV; 450 nm gives about 2.76 eV ($4.42 \times 10^{-19}$ J).

**Trick:** When $K_{max}$ is in eV, the stopping potential in volts has the same numerical value.

# 25.2 Wave-Particle Duality

## DEFINITIONS & LAWS

- **Wave-particle duality**: radiation and matter exhibit both wave properties (diffraction, interference) and particle properties (photoelectric effect, photon momentum) depending on the experiment.
- **de Broglie hypothesis**: a moving particle of momentum p has an associated wavelength $\lambda = \frac{h}{p}$.
- Wave nature of electrons is shown by electron diffraction through a thin crystal or graphite film (Davisson–Germer experiment): a ring (diffraction) pattern is formed.
- Particle nature of light is shown by the photoelectric effect; wave nature of light by diffraction and interference.
- The wave behaviour of macroscopic objects is not observed because the de Broglie wavelength is far too small compared with any obstacle or gap.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **de Broglie wavelength** | $\lambda = \frac{h}{p} = \frac{h}{mv}$ | λ = de Broglie wavelength (m); h = Planck constant (J s); p = momentum (kg m $s^{-1}$); m = mass (kg); v = speed (m $s^{-1}$) |
| **Wavelength from kinetic energy** | $p = \sqrt{2mK}$; $\lambda = \frac{h}{\sqrt{2mK}}$ | K = kinetic energy (J); m = mass (kg); p = momentum (kg m $s^{-1}$); λ = wavelength (m) |
| **Electron accelerated through a potential difference** | $eV = \frac{1}{2}mv^2$; $\lambda = \frac{h}{\sqrt{2meV}}$ | V = accelerating potential difference (V); e = electron charge (C); m = electron mass (kg); v = speed (m $s^{-1}$); λ = wavelength (m) |

- Constants: h = 6.63 × $10^{-34}$ J s, $m_e$ = 9.11 × $10^{-31}$ kg, e = 1.60 × $10^{-19}$ C, $m_p$ = 1.67 × $10^{-27}$ kg.

## TIPS TO ANSWER

- Write $\lambda = \frac{h}{mv}$ (or $\frac{h}{\sqrt{2meV}}$), substitute with SI units and give the answer in metres to 3 s.f. (2 s.f. if the data are 2 s.f.).
- For evidence of both natures name the experiment AND the property shown: diffraction/interference → wave; photoelectric effect → particle.
- For an electron accelerated through V, first find the speed or momentum from $eV = \frac{1}{2}mv^2$, then $\lambda = \frac{h}{p}$.
- State: a larger momentum gives a shorter wavelength ($\lambda \propto \frac{1}{p}$).

### SECTION A TRAPS

**Trap:** Saying that the photoelectric effect alone shows both natures: it demonstrates the particle nature only. Two different phenomena are needed.

**Trap:** Using the speed of light for the electron or confusing the mass of a proton ($1.67 \times 10^{-27}$ kg) with that of an electron. A proton at the same speed has a wavelength about 1800 times shorter.

**Trap:** Forgetting the square root: $\lambda = \frac{h}{\sqrt{2meV}}$, not $\frac{h}{2meV}$. For V = 150 V, $\lambda \approx 1.0 \times 10^{-10}$ m.

**Trap:** Rutherford scattering and the Millikan experiment do not show the wave nature of the electron.

### TRICKS

**Trick:** Fast electron wavelength: $\lambda$ (nm) $\approx \frac{1.23}{\sqrt{V}}$ (V in volts). For 150 V, $\lambda \approx 0.100$ nm.

**Trick:** Same speed, heavier particle: $\lambda \propto \frac{1}{m}$, so a proton's wavelength is $\frac{1}{1836}$ that of an electron.

## 25.3 Atomic Structure

### DEFINITIONS & LAWS

- **Line spectrum** consists of discrete bright lines (emission) or dark lines (absorption) of definite wavelengths, characteristic of the element; it shows that electrons in atoms occupy discrete energy levels.
- **Bohr's postulates**: electrons move in certain stable orbits without radiating energy (stationary states); the angular momentum of an orbiting electron is quantised, $mvr = \frac{nh}{2\pi}$; radiation is emitted or absorbed only when an electron makes a transition between two levels, $hf = E_i - E_f$.
- **Energy level** is the energy of an electron in an allowed orbit in the atom.
- **Ground state** is the state in which the electron in the atom has the lowest energy (n = 1); an **excited state** is any state with energy higher than the ground state.
- **Excitation energy** is the energy needed to raise an electron from the ground state to an excited state.
- **Ionisation energy** is the minimum energy needed to remove an electron from the ground state of an atom so that it is completely detached from the atom ($n = \infty$); for hydrogen it is 13.6 eV.
- **Spectral series of hydrogen**: Lyman (transitions to n = 1, ultraviolet), Balmer (to n = 2, visible), Paschen (to n = 3, infrared).

- **Laser action**: stimulated emission produces photons identical in phase, frequency and direction to the stimulating photon; population inversion is needed for light amplification.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Hydrogen energy levels** | $E_n = -\frac{13.6}{n^2}\ \text{eV}$ | $n = 1, 2, 3, \ldots$ ($n = 1$ is the ground state); $E_n$ = energy of level n (eV); 1 eV = $1.60 \times 10^{-19}$ J |
| **Photon emitted or absorbed in a transition** | $hf = \frac{hc}{\lambda} = E_i - E_f$ | $E_i$ = energy of the initial (higher) level (J); $E_f$ = energy of the final level (J); f = photon frequency (Hz); $\lambda$ = wavelength (m) |
| **Rydberg formula** | $\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$ | R = $1.10 \times 10^7\ \text{m}^{-1}$ (Rydberg constant); $\lambda$ = wavelength (m); $n_i$ = initial level; $n_f$ = final level |
| **Bohr radius of orbit** | $r_n = n^2 a_0$ | $r_n$ = radius of the nth orbit (m); $a_0 = 5.29 \times 10^{-11}$ m (Bohr radius); $n = 1, 2, 3, \ldots$ |

## TIPS TO ANSWER

- Write the energies of both levels using $-\frac{13.6}{n^2}$, subtract (higher − lower) to obtain a positive photon energy, convert eV to J, then $\lambda = \frac{hc}{E}$.
- Explain line spectra in 2 points: electrons have only discrete energy levels; photons emitted in transitions have energy equal to the difference of two levels, so only specific wavelengths appear.
- Name the series (Lyman/Balmer/Paschen) and the region of the spectrum when asked.
- Give the wavelength in nm to 3 s.f. and the energy in eV.

## SECTION A TRAPS

**Trap:** Energy levels are negative; n = 2 (−3.40 eV) is higher than n = 1 (−13.6 eV). The photon energy for n = 2 → 1 is 10.2 eV, not 3.40 eV (which is the energy of the n = 2 level itself) or 13.6 eV (the ionisation energy).

**Trap:** Mistaking the Balmer wavelengths: n = 3 → 2 is 658 nm (red) and n = 5 → 2 is 435 nm (violet); 122 nm is the n = 2 → 1 (Lyman) line, in the ultraviolet.

**Trap:** Writing 'electrons move in continuous orbits so any wavelength is emitted'; the quantisation of energy levels is the point.

**Trap:** Dividing the photon energy in eV by hc directly without converting to joules.

### TRICKS

**Trick:** Shorter wavelength = larger energy gap. Lyman lines are shortest, then Balmer, then Paschen.

**Trick:** Minimum photon wavelength to ionise ground-state hydrogen: $\lambda = \frac{hc}{13.6}$ eV = 91 nm.

## 25.4 X-rays

### DEFINITIONS & LAWS

- **X-rays** are electromagnetic waves of very short wavelength (about $10^{-11}$ to $10^{-9}$ m), produced when fast electrons are suddenly stopped by a metal target.
- **Continuous spectrum** arises from the deceleration of electrons in the target (bremsstrahlung); it has a sharp cut-off at the minimum wavelength $\lambda_{min}$, corresponding to the whole kinetic energy of an electron being converted to a single photon.
- **Characteristic spectrum** consists of sharp lines produced when an incident electron ejects an inner-shell electron of the target atom and an outer electron falls to fill the vacancy; the lines are characteristic of the target material.
- **Bragg's law**: constructive interference occurs for X-rays reflected from successive crystal planes when the path difference 2d sin θ is a whole number of wavelengths.
- **Glancing angle** θ is the angle between the incident X-ray beam and the atomic (crystal) plane.
- Intensity of the X-ray beam depends on the tube current; the penetrating power (hardness) depends on the accelerating voltage.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Kinetic energy of electron at target** | $eV = \frac{1}{2}mv^2$ | electron reaching the target from rest; V = accelerating potential difference (V); e = electron charge (C); m = electron mass (kg); v = speed at the target (m $s^{-1}$) |
| **Minimum wavelength** | $eV = \frac{hc}{\lambda_{min}}$; $\lambda_{min} = \frac{hc}{eV}$ | $\lambda_{min}$ = minimum wavelength (m); h = 6.63 × $10^{-34}$ J s; c = 3.00 × $10^8$ m $s^{-1}$; e = 1.60 × $10^{-19}$ C; V = accelerating potential difference (V) |
| **Bragg's law** | $2d \sin\theta = n\lambda$ | d = spacing of the crystal planes (m); θ = glancing angle (measured from the planes, not the normal); n = 1, 2, 3, … (order); λ = wavelength (m) |
| **Attenuation of X-rays** | $I = I_0 e^{-\mu x}$ | $I_0$ = incident intensity (W $m^{-2}$); I = transmitted intensity (W $m^{-2}$); μ = linear attenuation coefficient ($m^{-1}$); x = thickness of the absorber (m) |

### TIPS TO ANSWER

- For $\lambda_{min}$ write $\lambda_{min} = \frac{hc}{eV}$, substitute with units; state that $\lambda_{min}$ depends only on V and not on the target material.
- For Bragg's law, state that θ is the glancing angle between the beam and the crystal planes; write $2d \sin \theta = n\lambda$ and solve.
- For the characteristic lines, state 'transition of an outer electron to a vacancy in an inner shell'.
- When asked for the electron speed, equate eV to $\frac{1}{2}mv^2$ and convert kV to V.

### SECTION A TRAPS

**Trap:** Using V in kV directly: 25 kV = $2.5 \times 10^4$ V. Forgetting this gives $\lambda_{min}$ a factor of 1000 error (options such as $3.1 \times 10^{-10}$ m for 40 kV).

**Trap:** Doubling V halves $\lambda_{min}$ ($\lambda_{min} \propto \frac{1}{V}$); options that double $\lambda_{min}$ or leave it unchanged are wrong.

**Trap:** Using the angle from the normal in Bragg's law: θ is the glancing angle measured from the crystal planes.

**Trap:** Saying the continuous spectrum is characteristic of the target; only the sharp lines depend on the target material.

### TRICKS

**Trick:** Quick estimate: $\lambda_{min}$ (nm) ≈ $\frac{1.24}{V}$ (kV). For 25 kV, $\frac{1.24}{25}$ = 0.0496 nm = $5.0 \times 10^{-11}$ m.

**Trick:** Bragg: $\sin \theta \leq 1$, so the wavelength must satisfy $\lambda \leq \frac{2d}{n}$ for order n.

## 25.5 Nanoscience

### DEFINITIONS & LAWS

- **Nanoscience** is the study of materials and structures with at least one dimension of the order of 1 nm to 100 nm.
- The surface-area-to-volume ratio of nanoparticles is very large, so a large proportion of atoms lie at the surface, which gives a high chemical reactivity and catalytic activity.

- **Scanning tunnelling microscope (STM)** uses the quantum-mechanical tunnelling current between a sharp conducting tip and the surface; the current depends very sensitively (exponentially) on the tip–surface separation, so atoms can be imaged.
- Examples: carbon nanotubes (high strength, electrical conduction), graphene, nanoparticles in sunscreens and catalysts.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Surface area to volume ratio of a sphere** | $\frac{3}{r}$ | r = radius of the sphere (m); ratio in $m^{-1}$; the ratio increases as the size decreases |
| **Surface area to volume ratio of a cube** | $\frac{6}{L}$ | L = side of the cube (m); ratio in $m^{-1}$; the ratio increases as the size decreases |

- 1 nm = $10^{-9}$ m; the nanoscale is about 1–100 nm.

## TIPS TO ANSWER

- For reactivity state: smaller particles have a larger surface-area-to-volume ratio, so more atoms are exposed at the surface.
- For STM state: tunnelling current → very sensitive to the separation → used to map the surface atom by atom.
- Give the nanoscale range as 1 nm to 100 nm.

## SECTION A TRAPS

**Trap:** Quoting the nanoscale as 1 µm to 100 µm or 1 pm to 100 pm; the nanoscale is 1–100 nm.

**Trap:** Saying that nanoparticles obey different chemistry or have fewer atoms 'so each atom is more reactive'; the reason is the large surface-area-to-volume ratio.

## TRICKS

**Trick:** For surface-area-to-volume ratio: halving the radius doubles the ratio $\frac{3}{r}$.

Chapter 26

# NUCLEAR PHYSICS

## 26.1 Nucleus

### DEFINITIONS & LAWS

- The **proton (atomic) number** Z is the number of protons in the nucleus; the **nucleon (mass) number** A is the total number of protons and neutrons; the number of neutrons N = A − Z.
- **Isotopes** are nuclides of the same element (same Z) with different numbers of neutrons (different A).
- A **nuclide** is a particular species of nucleus with a specified proton number Z and nucleon number A.
- The **unified atomic mass unit** (u) is one twelfth of the mass of a neutral carbon-12 atom; 1 u = $1.66\times10^{-27}$ kg ≡ 931.5 MeV.
- The **mass defect** is the difference between the total mass of the separate nucleons and the mass of the nucleus.
- The **binding energy** is the energy required to separate a nucleus into its individual nucleons; $B = \Delta mc^2$.
- The **binding energy per nucleon** is the binding energy of a nucleus divided by its number of nucleons. It is a measure of nuclear stability: a nucleus with a greater binding energy per nucleon is more stable; the curve rises steeply for light nuclei, peaks at about 8.8 MeV near $^{56}Fe$ (A ≈ 56) and falls slowly for heavy nuclei.
- The nuclear radius $R = R_0A^{1/3}$, so the nuclear density is the same for all nuclei (about $2\times10^{17}$ kg m$^{-3}$).

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Mass defect** | $\Delta m = Zm_p + (A - Z)m_n - m_{nucleus}$ | Δm = mass defect (u); Z = proton number; A = nucleon number; $m_p$, $m_n$ = masses of proton and neutron (u); $m_{nucleus}$ = mass of nucleus (u); use atomic masses with $m_H$ when electrons are included |
| **Binding energy** | $B = \Delta mc^2$; binding energy per nucleon = $\frac{B}{A}$ | B = binding energy; Δm = mass defect (u); c = speed of light; 1 u = 931.5 MeV; A = nucleon number |
| **Nuclear radius** | $R = R_0A^{1/3}$ | R = nuclear radius (m); $R_0 \approx 1.2\times10^{-15}$ m (1.2 fm); A = nucleon number |
| **Nuclear density** | $\rho = Am_u/(\frac{4}{3}\pi R^3) = \frac{3m_u}{4\pi R_0{}^3}$ | ρ = density (kg m$^{-3}$); $m_u$ = atomic mass unit (kg); R = nuclear radius (m); $R_0 = 1.2\times10^{-15}$ m; independent of A |

- 1 MeV = $1.60\times10^{-13}$ J; c = $3.00\times10^8$ m s$^{-1}$.

### TIPS TO ANSWER

- For a nuclide, state the proton, neutron and nucleon numbers from the symbol $^{A}_{Z}X$; write $N = A - Z$ explicitly.
- For binding energy: $\Delta m \rightarrow B = \Delta mc^2$ (or $\times$ 931.5 MeV $u^{-1}$) $\rightarrow \frac{B}{A}$; show masses to enough decimal places.
- Use $R \propto A^{1/3}$ ratios; show the cube root clearly and state the ratio to 2 or 3 s.f.
- For the binding-energy curve: label axes ($\frac{B}{A}$ against A), mark the peak near $A \approx 56$, and state that fusion of light nuclei and fission of heavy nuclei both release energy because $\frac{B}{A}$ increases.

### SECTION A TRAPS

**Trap:** Taking neutron number as A instead of A − Z (e.g. 64 neutrons for $^{64}_{29}Cu$).

**Trap:** Quoting total binding energy when the question asks for binding energy per nucleon; divide by A.

**Trap:** Using $R \propto A$ instead of $A^{1/3}$; doubling A increases R by 1.26, not 2.

**Trap:** Placing the peak of the $\frac{B}{A}$ curve at helium or uranium; it is at iron ($A \approx 56$).

### TRICKS

**Trick:** Cube-root shortcuts: 8 → 2, 27 → 3, 64 → 4, 125 → 5, 216 → 6.

**Trick:** Density of the nucleus is independent of A: any option that depends on A is wrong.

## 26.2 Radioactivity

### DEFINITIONS & LAWS

- **Radioactive decay** is the spontaneous and random disintegration of an unstable nucleus with the emission of α, β or γ radiation.
- Radioactive decay is **random** and **spontaneous**. Random: it is impossible to predict when a particular nucleus will decay, and every nucleus has the same probability of decay per unit time. Spontaneous: the decay is not affected by external physical conditions such as temperature and pressure.

- The **activity** A is the rate of decay of the nuclei, $A = -\frac{dN}{dt} = \lambda N$, measured in becquerel (Bq; 1 Bq = 1 decay per second).
- The **decay constant** λ is the probability of decay per unit time of a nucleus.
- The **half-life** $t_{½}$ is the time taken for the number of undecayed nuclei (or the activity) to fall to half of its initial value.
- α emission reduces A by 4 and Z by 2; $\beta^-$ emission raises Z by 1 with A unchanged; γ emission changes neither A nor Z.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Decay law** | $N = N_0e^{-\lambda t}$; $A = A_0e^{-\lambda t}$ | N = number of undecayed nuclei; A = activity (Bq); $N_0$, $A_0$ = values at t = 0; λ = decay constant ($s^{-1}$); t = time (s) |
| **Half-life** | $t_{½} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$ | $t_{½}$ = half-life (s); λ = decay constant ($s^{-1}$) |
| **Half-life form of decay law** | $N = N_0(½)^{t/t_{½}}$; fraction remaining = $(½)^{t/t_{½}}$ | N = number of undecayed nuclei; $N_0$ = initial number; t = time (s); $t_{½}$ = half-life (s) |
| **Activity** | $A = \lambda N$; $A \propto N$ | A = activity (Bq); λ = decay constant ($s^{-1}$); N = number of undecayed nuclei |
| **Linearised decay law** | $\ln A = \ln A_0 - \lambda t$ | A = activity (Bq); $A_0$ = initial activity (Bq); λ = decay constant ($s^{-1}$); t = time (s); graph of ln A against t is a straight line with gradient −λ |

## TIPS TO ANSWER

- Convert the half-life to seconds when the activity is required in Bq: $t_{½}$ in s → $\lambda = \frac{\ln 2}{t_{½}}$ → $A = \lambda N$.
- State the number of half-lives $n = \frac{t}{t_{½}}$ first, then fraction = $(½)^n$; non-integer n is allowed with the exponential form.
- Show the substitution, the evaluated result to 3 s.f., and the unit (Bq, $s^{-1}$ or mass in the unit asked).
- For explanation questions, state 'random and spontaneous' and refer to constant probability of decay per unit time.

## SECTION A TRAPS

**Trap:** Taking fraction remaining after n half-lives as $\frac{1}{2n}$ or $1 - \frac{n}{2}$; the correct value is $(½)^n$.

**Trap:** Leaving $t_{½}$ in days or years when computing A in Bq; λ must be in $s^{-1}$.

**Trap:** Equating half-life with the time for all nuclei to decay, or half of the 'total life'. The half-life applies at any starting point.

**Trap:** Confusing $\beta^-$ with $\alpha$: $\beta^-$ increases Z by 1 and leaves A unchanged.

### TRICKS

**Trick:** Count half-lives: 28 days with $t_{½}$ = 8.0 days is 3.5 half-lives, so fraction = $(½)^{3.5} \approx 0.088$.

**Trick:** The mass of undecayed material and the activity fall by the same fraction as N.

## 26.3 Nuclear Reaction

### DEFINITIONS & LAWS

- In a nuclear reaction, mass number, charge (atomic number), energy and momentum are conserved; rest mass is not conserved.
- The **Q-value** of a reaction is the energy released (Q > 0) or absorbed (Q < 0), Q = (total mass of reactants − total mass of products)$c^2$.
- **Nuclear fission** is the splitting of a heavy nucleus (e.g. $^{235}U$, after absorbing a neutron) into two lighter nuclei with the release of energy and neutrons; each fission of $^{235}U$ releases about 200 MeV.
- **Nuclear fusion** is the joining of two light nuclei to form a heavier nucleus with the release of energy; it requires a very high temperature to overcome electrostatic repulsion.
- Energy is released in fission and fusion because the products have a greater binding energy per nucleon, so their total mass is less than that of the reactants.
- A **chain reaction** is sustained when, on average, at least one of the secondary neutrons from each fission causes another fission.
- A **thermal neutron** is a slow neutron whose kinetic energy is about the same as the thermal energy of particles at room temperature (about 0.025 eV); thermal neutrons are the most likely to cause fission of $^{235}U$.
- A **thermonuclear reaction** is a fusion reaction that occurs when the temperature of the light nuclei is raised to about $10^8$ K, so that they collide with enough kinetic energy to overcome the electrostatic repulsion and fuse.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Q-value of a reaction** | $Q = \Delta mc^2 = (\Sigma m_{reactants} - \Sigma m_{products}) \times 931.5$ MeV | Q = energy released (MeV); Δm = mass difference (u); masses in u; c = speed of light |

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Mass-energy equivalence** | $E = \Delta mc^2$ | E = energy (J); Δm = mass difference (kg); c = $3.00\times10^{8}$ m $s^{-1}$; 1 u = $1.66\times10^{-27}$ kg |
| **Nuclear equation** | ΣA and ΣZ equal on both sides; e.g. $^{2}H + {}^{3}H \rightarrow {}^{4}He + n$ | A = nucleon number; Z = proton number; both conserved in a nuclear reaction |
| **Power from fission** | P = (fission rate) × (energy per fission) | P = power (W); fission rate in $s^{-1}$; energy per fission in J |
| **Mass of $^{235}U$ used** | $\text{Mass} = \frac{\text{number of fissions} \times 235\ \text{g}}{N_A}$ | Mass in g; $N_A$ = Avogadro constant ($mol^{-1}$) |

- 1 eV = $1.60\times10^{-19}$ J; $N_A = 6.02\times10^{23}$ $mol^{-1}$.

## TIPS TO ANSWER

- Balance the nuclear equation first, checking the top (A) and bottom (Z) numbers separately.
- Calculate the mass defect with masses to 6 decimal places, then multiply by 931.5 MeV $u^{-1}$; give Q to 3 s.f. with unit, and state the sign.
- For power problems, convert MeV to J, then find the fission rate, then the amount of fuel consumed in the stated time (seconds in a day = 86 400).
- For a decay such as α emission, use the masses of the nuclei given; do not add electrons unless atomic masses are specified.

## SECTION A TRAPS

**Trap:** Forgetting to convert Δm to energy (leaving the answer in u) or multiplying by 931.5 when Δm is already in kg.

**Trap:** Subtracting reactants from products, giving a negative Q for an exothermic reaction.

**Trap:** Using the mass of the electron in nuclear masses inconsistently: atomic and nuclear masses must not be mixed.

**Trap:** Taking a molar mass of 235 kg $mol^{-1}$ instead of 0.235 kg $mol^{-1}$ (235 g $mol^{-1}$).

## TRICKS

**Trick:** If the reaction releases energy, total mass decreases: Δm = (before) − (after) must be positive.

> **Trick:** Energy check: 1 u ≈ 931.5 MeV, so Δm of about 0.02 u gives about 19 MeV: any option far from that order is wrong.